

Predictive Modeling of Concrete Compressive Strength and Curing Age
Kenshi King

Section 1: Descriptive Statistics

We begin our analysis by examining this data's characteristics via descriptive statistics. We make sure to examine these various statistics to ensure that the data meets statistical assumptions that allows us to then interpret the data with accuracy. Based on the Normality assessment, all continuous predictors yielded $p < 0.01$ on the formal normality tests, providing strong evidence against normality. The bar charts and probability plots show that the variables are all right-skewed, coinciding with our normality analysis. This analysis does make sense logically as concrete strength cannot be negative and a majority of the strengths are around a common range, typical strength level, with only a few high-strength outliers. For example, looking at fly ash and superplasticizer statistics, some observations have small amounts with others occasionally having higher amounts, resulting in producing the long right tails visualized in those variables graphics.

It is worth mentioning that with a sample size of this scale ($n \approx 1030$), formal normality tests are very sensitive and will reject normality even for negligible deviations; the histogram and probability plots will yield a more interpretable picture of the actual distributional strength.

Moments			
N	1030	Sum Weights	1030
Mean	1.57827483	Sum Observations	1625.62308
Std Deviation	0.64810507	Variance	0.42004019
Skewness	0.81434411	Kurtosis	0.26211969
Uncorrected SS	2997.90134	Corrected SS	432.221352
Coeff Variation	41.0641455	Std Error Mean	0.02019421

Tests for Normality			
Test	Statistic		p Value
Shapiro-Wilk	W	0.945224	Pr < W <0.0001
Kolmogorov-Smirnov	D	0.097957	Pr > D <0.0100
Cramer-von Mises	W-Sq	2.445746	Pr > W-Sq <0.0050
Anderson-Darling	A-Sq	15.45044	Pr > A-Sq <0.0050

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Section 1: Descriptive Statistics

The UNIVARIATE Procedure
Variable: slagwater

Moments			
N	1030	Sum Weights	1030
Mean	0.40685282	Sum Observations	419.0584
Std Deviation	0.47196048	Variance	0.22274669
Skewness	0.77757172	Kurtosis	-0.5883785
Uncorrected SS	399.701439	Corrected SS	229.206349
Coeff Variation	116.002756	Std Error Mean	0.01470574

Tests for Normality			
Test	Statistic		p Value
Shapiro-Wilk	W	0.81445	Pr < W <0.0001
Kolmogorov-Smirnov	D	0.26288	Pr > D <0.0100
Cramer-von Mises	W-Sq	13.5408	Pr > W-Sq <0.0050
Anderson-Darling	A-Sq	78.38401	Pr > A-Sq <0.0050

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Section 1: Descriptive Statistics

The UNIVARIATE Procedure
Variable: flyashwater

Moments			
N	1030	Sum Weights	1030
Mean	0.31341707	Sum Observations	322.819579
Std Deviation	0.3756394	Variance	0.14110496
Skewness	0.63080779	Kurtosis	-1.0655919
Uncorrected SS	246.37417	Corrected SS	145.197005
Coeff Variation	119.852887	Std Error Mean	0.01170449

Tests for Normality			
Test	Statistic		p Value
Shapiro-Wilk	W	0.771123	Pr < W <0.0001
Kolmogorov-Smirnov	D	0.347474	Pr > D <0.0100
Cramer-von Mises	W-Sq	19.46315	Pr > W-Sq <0.0050
Anderson-Darling	A-Sq	108.6461	Pr > A-Sq <0.0050

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Section 1: Descriptive Statistics

The UNIVARIATE Procedure
Variable: superplasticizerwater

Moments			
N	1030	Sum Weights	1030
Mean	0.0374018	Sum Observations	38.5238507
Std Deviation	0.03913086	Variance	0.00153122
Skewness	1.45900882	Kurtosis	3.86341419
Uncorrected SS	3.01649123	Corrected SS	1.57562999
Coeff Variation	104.622951	Std Error Mean	0.00121927

Tests for Normality			
Test	Statistic		p Value
Shapiro-Wilk	W	0.837675	Pr < W <0.0001
Kolmogorov-Smirnov	D	0.198378	Pr > D <0.0100
Cramer-von Mises	W-Sq	5.169731	Pr > W-Sq <0.0050
Anderson-Darling	A-Sq	38.39208	Pr > A-Sq <0.0050

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Section 1: Descriptive Statistics

The UNIVARIATE Procedure
Variable: coarsewater

Moments			
N	1030	Sum Weights	1030
Mean	5.44318093	Sum Observations	5606.47636
Std Deviation	0.84296582	Variance	0.71059137
Skewness	0.42993385	Kurtosis	0.44606419
Uncorrected SS	31248.2637	Corrected SS	731.198523
Coeff Variation	15.4866397	Std Error Mean	0.02626584

Tests for Normality			
Test	Statistic		p Value
Shapiro-Wilk	W	0.979456	Pr < W <0.0001
Kolmogorov-Smirnov	D	0.05558	Pr > D <0.0100
Cramer-von Mises	W-Sq	0.432191	Pr > W-Sq <0.0050
Anderson-Darling	A-Sq	3.438589	Pr > A-Sq <0.0050

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Section 1: Descriptive Statistics

The UNIVARIATE Procedure
Variable: finewater

Moments			
N	1030	Sum Weights	1030
Mean	4.34476279	Sum Observations	4475.10567
Std Deviation	0.82490823	Variance	0.68047359
Skewness	0.48490253	Kurtosis	1.1551481
Uncorrected SS	20143.4799	Corrected SS	700.207319
Coeff Variation	18.9862662	Std Error Mean	0.02570319

Tests for Normality			
Test	Statistic		p Value
Shapiro-Wilk	W	0.973242	Pr < W <0.0001
Kolmogorov-Smirnov	D	0.088691	Pr > D <0.0100
Cramer-von Mises	W-Sq	1.406588	Pr > W-Sq <0.0050
Anderson-Darling	A-Sq	7.231088	Pr > A-Sq <0.0050

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Section 1: Descriptive Statistics

The UNIVARIATE Procedure
Variable: age

Moments			
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Moments			
N	1030	Sum Weights	1030
Mean	45.6621359	Sum Observations	47032
Std Deviation	63.1699116	Variance	3990.43773
Skewness	3.2691774	Kurtosis	12.168989
Uncorrected SS	6253742	Corrected SS	4106160.42
Coeff Variation	138.341999	Std Error Mean	1.96830165

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.590706	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.337291	Pr > D	<0.0100
Cramer-von Mises	W-Sq	24.34705	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	128.9549	Pr > A-Sq	<0.0050

Predictive Modeling of Concrete Compressive Strength and Curing Age

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Section 1: Descriptive Statistics

The UNIVARIATE Procedure
Variable: compressivestrength

Moments			
N	1030	Sum Weights	1030
Mean	35.8178358	Sum Observations	36892.3709
Std Deviation	16.7056792	Variance	279.079717
Skewness	0.41692228	Kurtosis	-0.3138437
Uncorrected SS	1608577.91	Corrected SS	287173.028
Coeff Variation	46.6406716	Std Error Mean	0.52052971

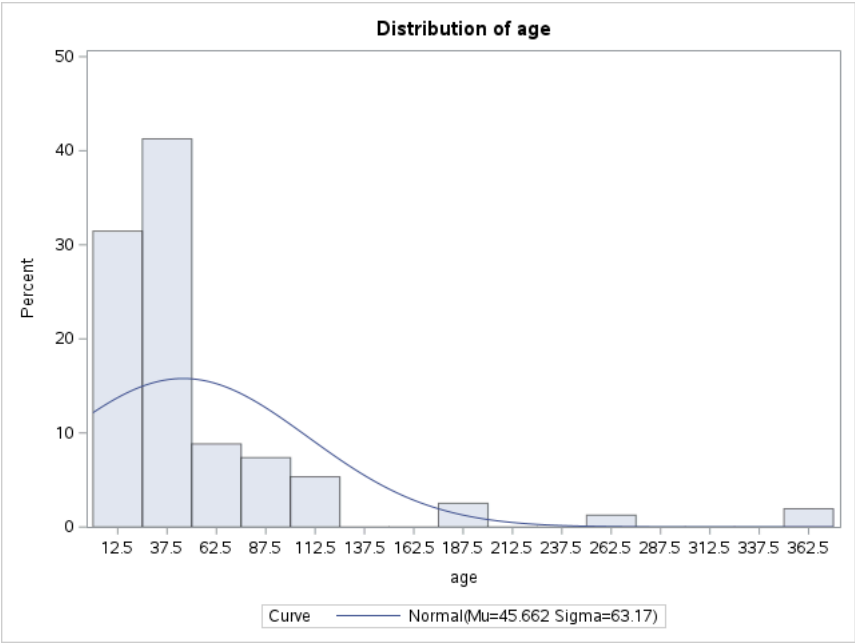
Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.979793	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.041303	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.520665	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	4.068344	Pr > A-Sq	<0.0050

Predictive Modeling of Concrete Compressive Strength and Curing Age

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Section 1: Descriptive Statistics

The UNIVARIATE Procedure

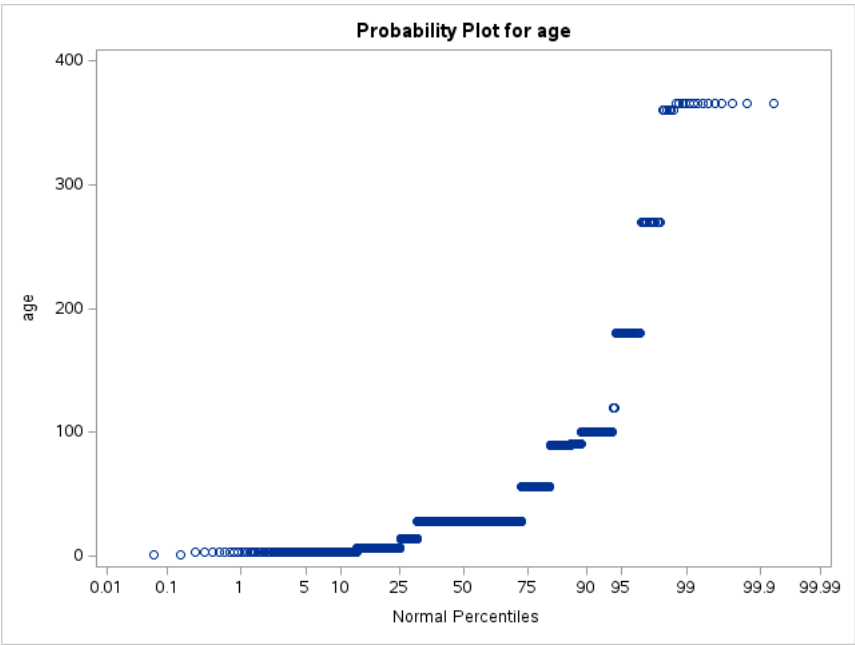


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Section 1: Descriptive Statistics

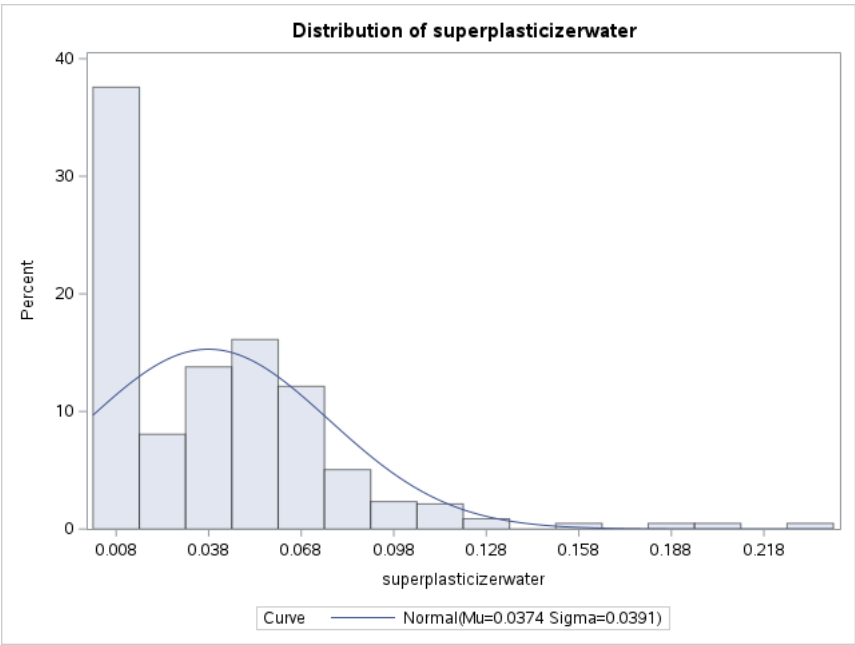
The UNIVARIATE Procedure



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Section 1: Descriptive Statistics

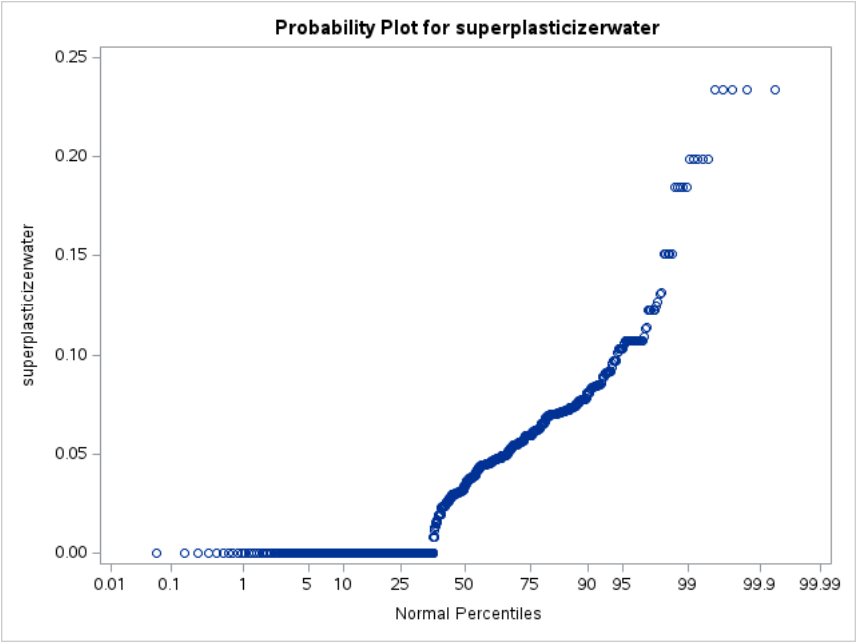
The UNIVARIATE Procedure



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Section 1: Descriptive Statistics

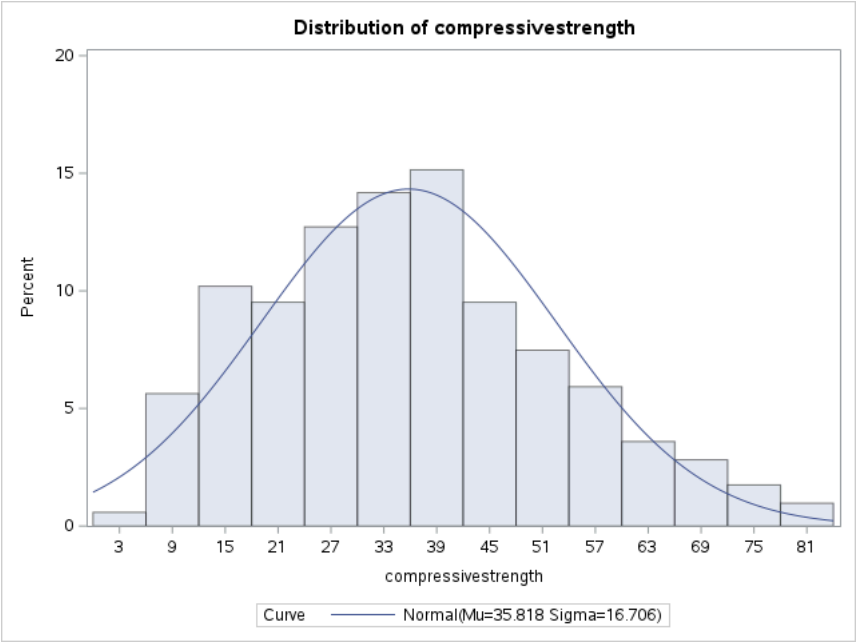
The UNIVARIATE Procedure



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Section 1: Descriptive Statistics

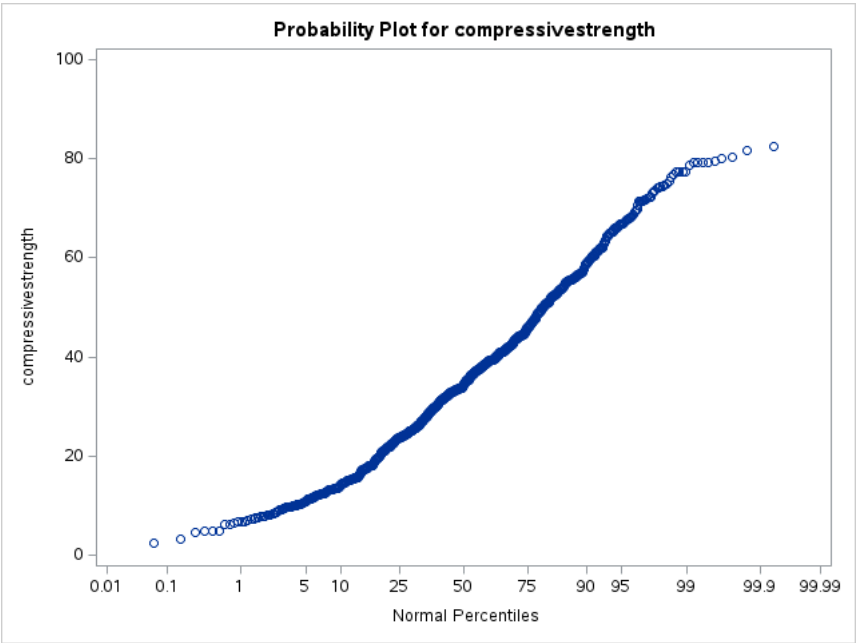
The UNIVARIATE Procedure



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The UNIVARIATE Procedure

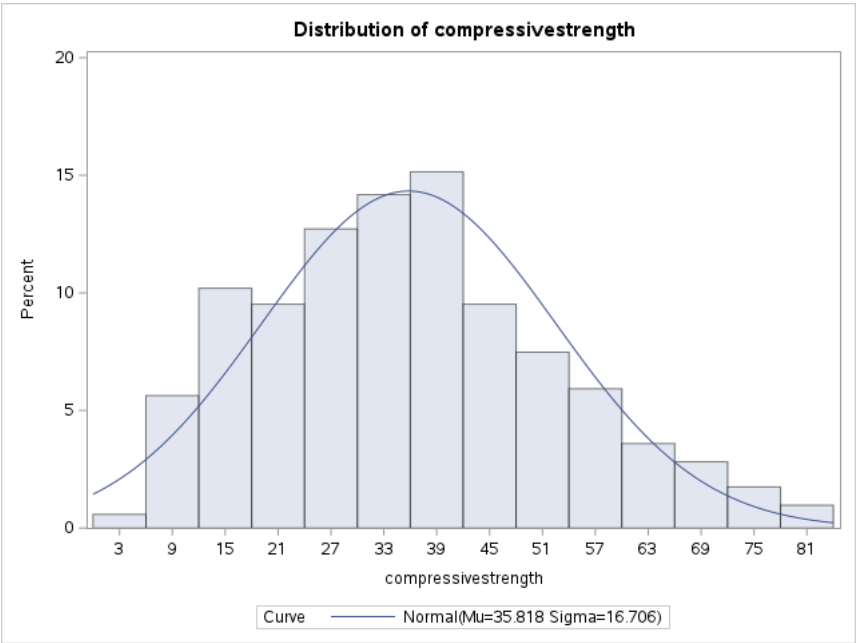


Predictive Modeling of Concrete Compressive Strength and Curing Age

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Section 1: Descriptive Statistics

The UNIVARIATE Procedure



The Spearman correlation analysis revealed insight into how the component-to-water ratios, age, and compressive strength relate to one another. As the aforementioned Abrams' Law states, the cement-to-water-ratio is strongly positively correlated with compressive strength ($p \sim 0.52$, $p < 0.0001$), while age shows an even stronger steady relationship ($p \sim 0.60$, $p < 0.0001$), confirming that both mixture composition and curing time are major contenders of concrete performance. The remaining component ratios display only weak to moderate correlations with strength, suggesting their effects are less influential predictors. Moderate positive correlations appear among several mix ratios. For instance, coarsewater and finewater ($p \sim 0.62$) and superplasticizerwater with other components (p ranging from ~ 0.47 to ~ 0.58). These relationships are expected since certain components tend to be modified together in concrete production. Importantly, no pair of predictors exceeds an absolute correlation of 0.7, indicating no immediate signs of multicollinearity. Additionally, age is nearly uncorrelated with components ratios ($|p| < 0.09$), suggesting that curing time is largely independent of mix composition. Overall, the correlation pattern highlights cement content and age as the most influential variables for strength while showing that the set of predictors maintain a reasonable level of independence.

Predictive Modeling of Concrete Compressive Strength and Curing Age

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Section 1: Descriptive Statistics

The CORR Procedure

Spearman Correlation Coefficients, N = 1030								
Prob > r under H0: Rho=0								
	cementwater	slagwater	flyashwater	superplasticizerwater	coarsewater	finewater	age	compressivestrength
cementwater	1.00000	-0.20064 <.0001	-0.28986 <.0001	0.24141 <.0001	0.25225 <.0001	0.15065 <.0001	-0.02823 0.3654	0.52190 <.0001

Spearman Correlation Coefficients, N = 1030								
Prob > r under H0: Rho=0								
	cementwater	slagwater	flyashwater	superplasticizerwater	coarsewater	finewater	age	compressivestrength
slagwater	-0.20064 <.0001	1.00000	-0.25956 <.0001	0.13814 <.0001	-0.17852 <.0001	-0.14236 <.0001	-0.02438 0.4345	0.19235 <.0001
flyashwater	-0.28986 <.0001	-0.25956 <.0001	1.00000	0.49200 <.0001	0.29368 <.0001	0.30711 <.0001	0.00232 0.9408	-0.05821 0.0618
superplasticizerwater	0.24141 <.0001	0.13814 <.0001	0.49200 <.0001	1.00000	0.47318 <.0001	0.58176 <.0001	-0.01123 0.7189	0.36174 <.0001
coarsewater	0.25225 <.0001	-0.17852 <.0001	0.29368 <.0001	0.47318 <.0001	1.00000	0.61825 <.0001	-0.08813 0.0046	0.14697 <.0001
finewater	0.15065 <.0001	-0.14236 <.0001	0.30711 <.0001	0.58176 <.0001	0.61825 <.0001	1.00000	-0.07034 0.0240	0.08894 0.0043
age	-0.02823 0.3654	-0.02438 0.4345	0.00232 0.9408	-0.01123 0.7189	-0.08813 0.0046	-0.07034 0.0240	1.00000	0.59602 <.0001
compressivestrength	0.52190 <.0001	0.19235 <.0001	-0.05821 0.0618	0.36174 <.0001	0.14697 <.0001	0.08894 0.0043	0.59602 <.0001	1.00000

In order to examine how compressive strength develops over time, the concrete samples were split into 5 different age groups. As expected, compressive strength increases with curing age: the youngest concretes (<28 days, Group 1) have the lowest mean strength (23.54 MPa) and the largest variability (SD ~ 12.41 MPa), reflecting early-stages in strength development. Strength rises exceptionally in Group 2 (28-55 days, mean ~ 36.75 MPa). Group 3 (56-89 days), Groups 4 (90-179 days) and 5 (>= 180 days) maintain relatively high strengths (51.89 MPa, 48.24 MPa and 44.16 MPa, respectively) with slightly lower variability, indicating that concrete strength stabilizes as it cures longer. The pattern confirms that both curing time and mixture composition strongly influence compressive strength, with strength increasing rapidly in early ages and leveling in older samples.

Note: Age groupings used for this descriptive analysis differ from the specific age thresholds later used in Section 3 (e.g., >= 100 days, 90-100 days), which were defined for a separate task.

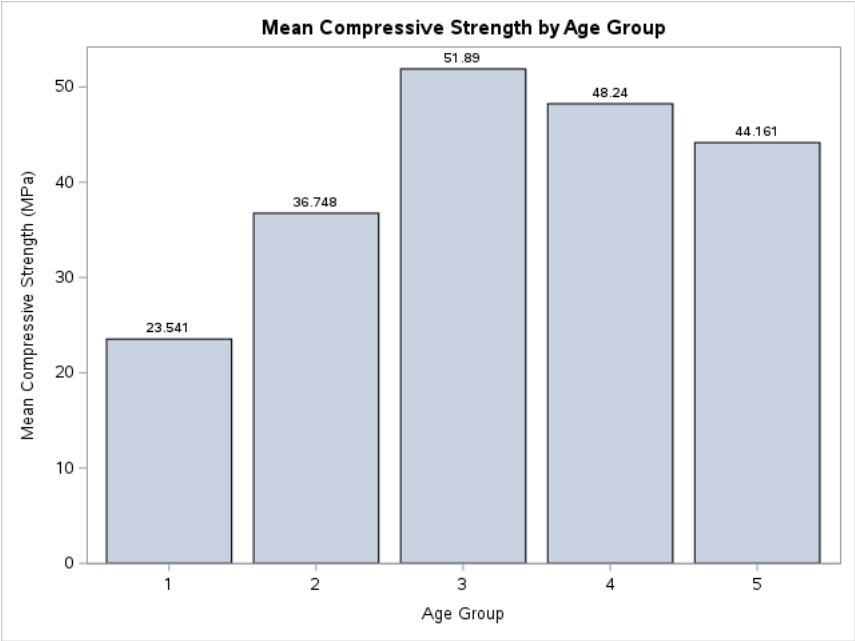
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Section 1: Descriptive Statistics

The MEANS Procedure

Analysis Variable : compressivestrength				
agegroup	N Obs	N	Mean	Std Dev
1	324	324	23.54	12.41
2	425	425	36.75	14.71
3	91	91	51.89	14.31
4	131	131	48.24	13.50
5	59	59	44.16	10.61



To explore natural groupings of concrete compositions and ages, hierarchical clustering was performed using the component-to-water ratios and age. Two procedural choices are worth noting: all variables were standardized prior to clustering, since the ratios and age are measured on very different metric scales and would otherwise let age dominate the grouping due to its larger numeric range; and compressive strength was deliberately withheld from the clustering variables themselves. The variable added back only afterward to describe each group, so that the resulting clusters represent a genuine, independent pattern in mix composition and curing time rather than one that superficially 'explains' strength by construction.

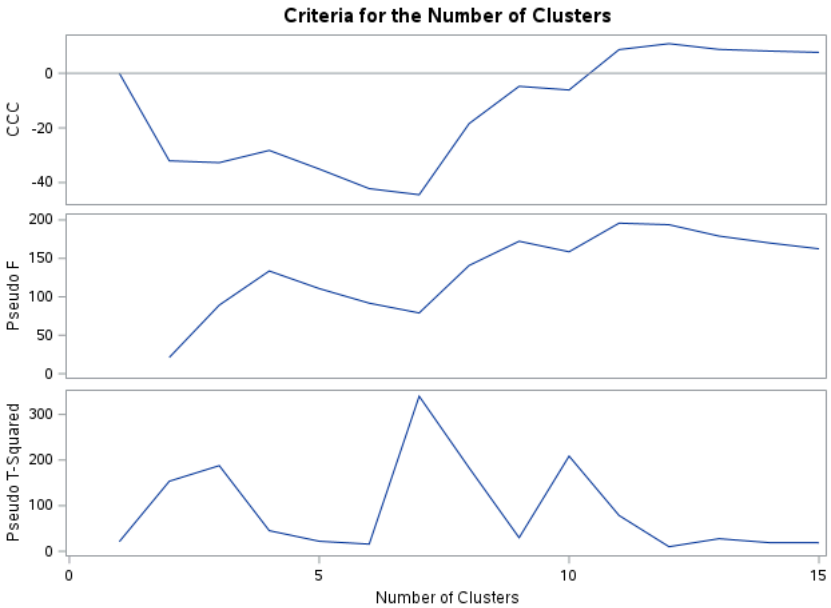
Based on the pseudo F statistics, the analysis peaked at four clusters (134, compared to 89.2 at three clusters), and the pseudo t^2 statistic jumped sharply when merging from four to three clusters. These support a four-cluster solution as the best-choice grouping. Cluster 1 (n=73) represents a high-performing mix profile, combining high cement and slag content with minimal fly ash, achieving the greatest average compressive strength (55.46 MPa) at a curing age (~36 days) comparable to the much larger dominant cluster. Cluster 2 (n=904) represents standard concrete mixes with lower cement content and moderate fly ash, averaging at a considerably lower strength (33.71 MPa) at a similar age. Cluster 3 (n=5) describes a more peculiar and small mixture with very high fly ash and aggregate content, as well as the lowest strength (24.38 MPa). Given its very small size, this probably reflects a handful of observations that are outliers rather than a specific, rare mixing strategy. Cluster 4 (n=48) is described by having high cement content, minimal additives, and a very long curing period (~281 days), achieving the greatest overall strength (46.75 MPa). These results suggest that composition, not just age, distinguishes great and poor-performing mixes even within similar curing periods. With regards to efficiency, cluster 1 in particular represents a proven strategy for achieving high early-to-moderate strength without the need of prolonged curing time.

Mean Compressive Strength by Age Group

Section 2: Clustering

The CLUSTER Procedure
Average Linkage Cluster Analysis

Cluster History											
Number of Clusters	Clusters Joined		Freq	Semipartial R-Square	R-Square	Approximate Expected R-Square	Cubic Clustering Criterion	Pseudo F Statistic	Pseudo t-Squared	Norm RMS Distance	Tie
15	CL177	CL31	15	0.0030	.691	.662	7.64	162	18.6	0.7631	
14	CL17	CL42	228	0.0062	.685	.653	8.12	170	19.0	0.7683	
13	CL27	CL16	53	0.0065	.679	.644	8.72	179	27.6	0.7776	
12	CL35	CL160	22	0.0019	.677	.634	10.8	194	10.0	0.809	
11	CL22	CL39	224	0.0191	.658	.623	8.65	196	78.5	0.8211	
10	CL18	CL14	456	0.0745	.583	.610	-6.1	159	209	0.8379	
9	CL12	CL20	42	0.0088	.574	.596	-4.8	172	29.8	0.8485	
8	CL11	CL10	680	0.0835	.491	.578	-18	141	183	0.8489	
7	CL8	CL21	904	0.1743	.317	.558	-44	79.0	340	0.9948	
6	CL9	CL64	48	0.0073	.309	.534	-42	91.8	15.7	1.0139	
5	CL25	CL13	58	0.0077	.302	.502	-35	111	22.0	1.0799	
4	CL5	CL15	73	0.0207	.281	.458	-28	134	45.2	1.1462	
3	CL7	CL4	977	0.1329	.148	.384	-33	89.2	188	1.3232	
2	CL3	CL6	1025	0.1279	.020	.276	-32	21.1	154	1.4756	
1	CL2	CL120	1030	0.0201	.000	.000	0.00	.	21.1	1.6111	



Mean Compressive Strength by Age Group

Section 2: Clustering

The CONTENTS Procedure

Data Set Name	WORK.DUPLICATES	Observations	34
Member Type	DATA	Variables	9
Engine	V9	Indexes	0
Created	09/14/2026 08:45:59	Observation Length	72
Last Modified	09/14/2026 08:45:59	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	YES
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	1816
Obs in First Data Page	34
Number of Data Set Repairs	0
Filename	/saswork/SAS_workB748000D89B_odaws01-usw2-2.oda.sas.com/SAS_workF8BE000D89B_odaws01-usw2-2.oda.sas.com/duplicates.sas7bdat
Release Created	9.0401M8
Host Created	Linux

Engine/Host Dependent Information	
Inode Number	2080375292
Access Permission	rw-r--r--
Owner Name	u64323947
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes			
#	Variable	Type	Len
7	age	Num	8
9	agegroup	Num	8
1	cementwater	Num	8
5	coarsewater	Num	8
8	compressivestrength	Num	8
6	finewater	Num	8
3	flyashwater	Num	8
2	slagwater	Num	8
4	superplasticizerwater	Num	8

Sort Information	
Sortedby	cementwater slagwater flyashwater superplasticizerwater coarsewater finewater age
Validated	YES
Character Set	ASCII

When running the program, a great number of tied minimum distances were flagged during the clustering process (389 under average linkage). Investigation confirmed only a small number of exact duplicate observations (34 out of ≈1,030) among the clustering variables, suggesting the ties are a normal consequence of how the clustering method handles many closely related mixtures, not a sign of a problem with the data itself.

Mean Compressive Strength by Age Group

Section 2: Clustering

The MEANS Procedure

CLUSTER=1

Variable	N	Mean	Std Dev
cementwater	73	2.77	0.48
slagwater	73	0.77	0.41
flyashwater	73	0.01	0.05
superplasticizerwater	73	0.12	0.06
coarsewater	73	6.26	0.71
finewater	73	5.76	0.81
age	73	35.73	31.56
compressivestrength	73	55.46	14.65

CLUSTER=2

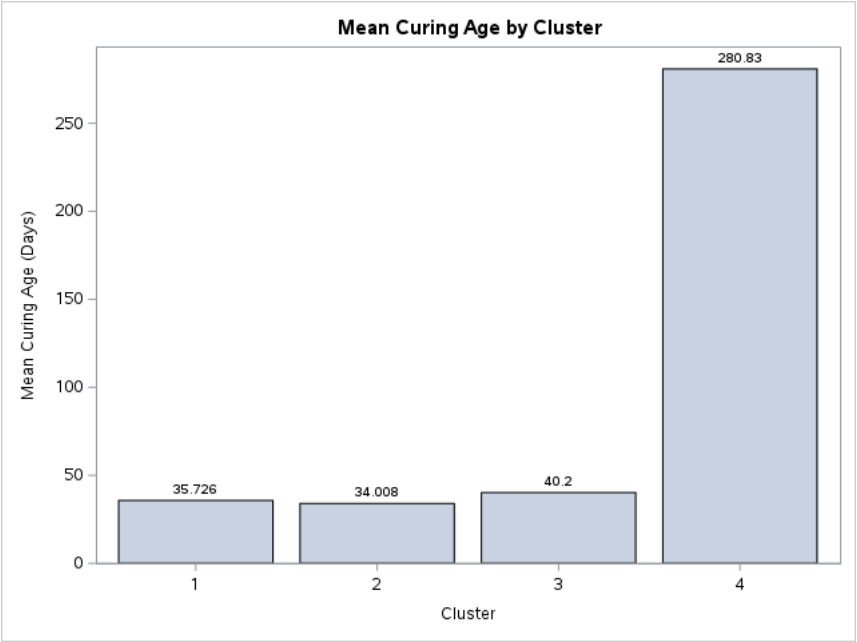
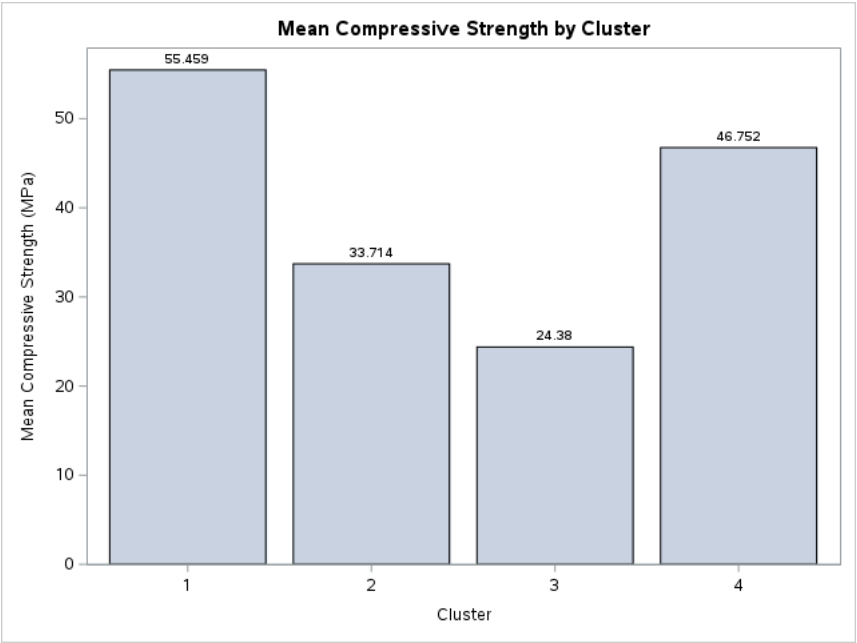
Variable	N	Mean	Std Dev
cementwater	904	1.48	0.56
slagwater	904	0.38	0.47
flyashwater	904	0.35	0.37
superplasticizerwater	904	0.03	0.03
coarsewater	904	5.40	0.77
finewater	904	4.28	0.66
age	904	34.01	32.48
compressivestrength	904	33.71	15.89

CLUSTER=3

Variable	N	Mean	Std Dev
cementwater	5	1.38	0.00
slagwater	5	0.35	0.00
flyashwater	5	1.35	0.00
superplasticizerwater	5	0.05	0.00
coarsewater	5	8.70	0.00
finewater	5	6.41	0.00
age	5	40.20	38.87
compressivestrength	5	24.38	12.37

CLUSTER=4

Variable	N	Mean	Std Dev
cementwater	48	1.60	0.58
slagwater	48	0.33	0.37
flyashwater	48	0.00	0.00
superplasticizerwater	48	0.00	0.00
coarsewater	48	4.59	0.73
finewater	48	3.22	0.67
age	48	280.83	78.66
compressivestrength	48	46.75	9.79



In this section, we are tasked with building various predictive models to provide useful insights for concrete performance and aging. Specifically, we aim to 1) predict the compressive strength of concrete that has cured for at least 100 days, 2) determine the likelihood that concrete cured between 90 and 100 days will achieve a strength of at least 50 MPa (for safety), and 3) estimate the approximate age of concrete samples based on their composition and observed strength data. These models will help the construction manager make informed decisions about concrete quality, curing timelines, and project efficiency. We will also assess model performance, identify the most influential factors, and highlight any limitations in predicting specific outcomes; we will begin with part 1).

Because compressive strength is a continuous variable, we apply a multiple linear regression model in order to predict compressive strength for concrete cured at least 100 days. While Section 1 identified non-normality in the raw predictor variables, regression validity is dependent on the normality of the model residuals rather than the predictors themselves, so this is tested below.

Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The REG Procedure
Model: MODEL1
Dependent Variable: compressivestrength

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation
Intercept	1	21.99493	6.46747	3.40	0.0009	0
cementwater	1	13.83188	2.96880	4.66	<.0001	4.56003
slagwater	1	14.24446	3.41561	4.17	<.0001	2.96130
flyashwater	1	3.98048	4.19714	0.95	0.3451	7.22409
superplasticizerwater	1	217.22744	46.35140	4.69	<.0001	4.54073
coarsewater	1	1.67997	1.98355	0.85	0.3989	10.56259
finewater	1	-3.94446	1.84767	-2.13	0.0351	8.64839

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation
age	1	0.00831	0.00896	0.93	0.3557	2.11687

When looking at the parameter estimates, Coarse-to-water ratio was not statistically significant in the full model (p = 0.399) and it showed an elevated variance inflation factor (VIF = 10.56). Given its lack of significance, it was removed from the model going forward, which is consistent with its exclusion in the subsequent stepwise-selected model in the next section.

Note: Superplasticizer-to-water ratio shows a considerably larger coefficient and standard error than the other predictors, indicating that its much more restricted measurement range rather than an extreme estimate. To further add, its t-value (4.69) confirms this remains one of the most accurate estimates and statistically significant predictor in the model.

Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The REG Procedure
Model: MODEL1
Dependent Variable: compressivestrength

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation
Intercept	1	22.14316	6.45656	3.43	0.0009	0
cementwater	1	15.73355	1.93970	8.11	<.0001	1.95174
slagwater	1	15.93063	2.77173	5.75	<.0001	1.95522
flyashwater	1	6.15763	3.31347	1.86	0.0659	4.51431
superplasticizerwater	1	213.53469	46.08489	4.63	<.0001	4.50055
finewater	1	-2.67807	1.08401	-2.47	0.0151	2.98469
age	1	0.00794	0.00894	0.89	0.3765	2.11171

When viewing the residuals from this model, significant disparity from normality was not present (Shapiro-Wilk p = 0.102; Kolmogorov-Smirnov, Cramer-von Mises, and Anderson-Darling tests all yielded p > 0.15), supporting the validity of this model's inference despite the skew present within Section 1.

Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

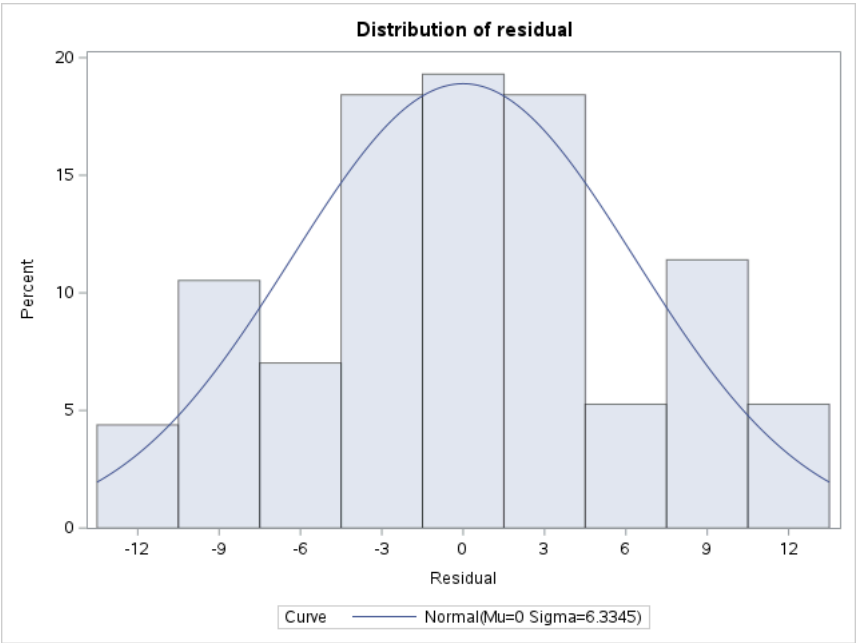
The UNIVARIATE Procedure
Variable: residual (Residual)

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.979066	Pr < W	0.0710
Kolmogorov-Smirnov	D	0.057832	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.073062	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.565845	Pr > A-Sq	0.1438

Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The UNIVARIATE Procedure



In order to assess how well this model generalizes beyond the data it was fit on, the sample was randomly split into a training set (70 and 80) and a test set (30 and 20, respectively). The stepwise regression below was fit on the training data, and then evaluted on the withheld test set.

Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling
------------------	------------------------

Input Data Set	CONCRETE100UP
Random Number Seed	777
Sampling Rate	0.7
Sample Size	80
Selection Probability	0.701754
Sampling Weight	0
Output Data Set	CONCRETE100UP_SAMPLE70

Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The REG Procedure
Model: MODEL1
Dependent Variable: compressivestrength

Number of Observations Read	80
Number of Observations Used	80

Stepwise Selection: Step 1

Variable superplasticizerwater Entered: R-Square = 0.1345 and C(p) = 70.4518

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1038.78074	1038.78074	12.12	0.0008
Error	78	6682.97009	85.67910		
Corrected Total	79	7721.75083			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	43.73690	1.30007	96970	1131.79	<.0001
superplasticizerwater	127.00271	36.47443	1038.78074	12.12	0.0008

Bounds on condition number: 1, 1

Stepwise Selection: Step 2

Variable finewater Entered: R-Square = 0.2932 and C(p) = 45.5945

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	2264.34715	1132.17357	15.97	<.0001
Error	77	5457.40368	70.87537		
Corrected Total	79	7721.75083			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	65.87736	5.45406	10340	145.89	<.0001
superplasticizerwater	292.41632	51.79636	2258.91593	31.87	<.0001
finewater	-6.38505	1.53548	1225.56640	17.29	<.0001

Bounds on condition number: 2.4378, 9.7513

Stepwise Selection: Step 3

Variable cementwater Entered: R-Square = 0.4069 and C(p) = 28.3696

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3141.62780	1047.20927	17.38	<.0001
Error	76	4580.12303	60.26478		
Corrected Total	79	7721.75083			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	53.15262	6.03461	4675.36043	77.58	<.0001
cementwater	7.80322	2.04520	877.28065	14.56	0.0003
superplasticizerwater	300.07077	47.80421	2374.53560	39.40	<.0001
finewater	-6.04627	1.41867	1094.65792	18.16	<.0001

Bounds on condition number: 2.4474, 17.74

Stepwise Selection: Step 4

Variable slagwater Entered: R-Square = 0.5407 and C(p) = 7.7178

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	4175.28633	1043.82158	22.07	<.0001
Error	75	3546.46450	47.28619		
Corrected Total	79	7721.75083			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	27.21399	7.70407	590.03638	12.48	0.0007
cementwater	15.15933	2.39948	1887.38948	39.91	<.0001
slagwater	15.70050	3.35809	1033.65853	21.86	<.0001
superplasticizerwater	273.83609	42.71510	1943.35868	41.10	<.0001
finewater	-3.01941	1.41361	215.73326	4.56	0.0359

Bounds on condition number: 3.097, 37.717

All variables left in the model are significant at the 0.0500 level.

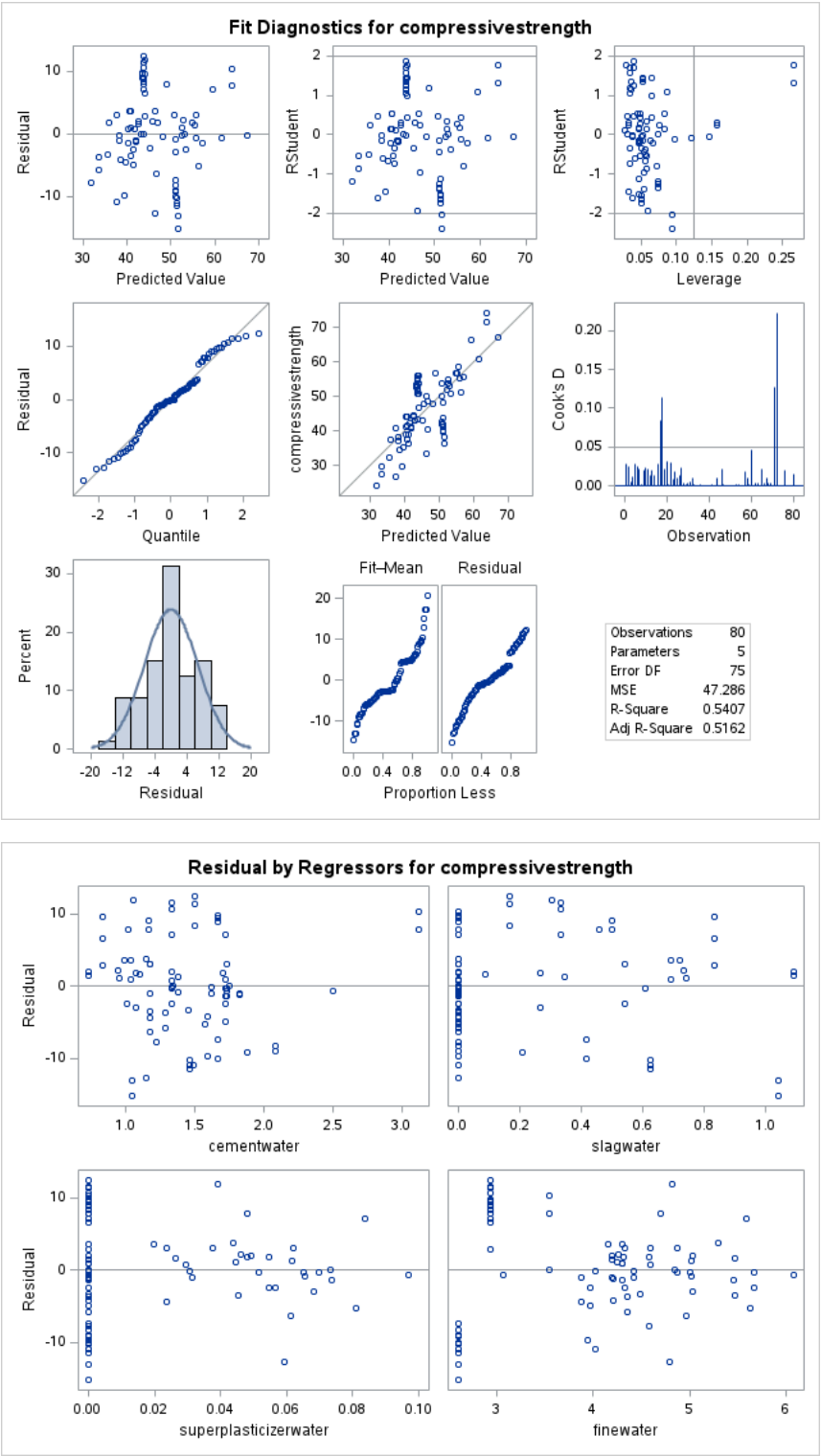
No other variable met the 0.0500 significance level for entry into the model.

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	superplasticizerwater		1	0.1345	0.1345	70.4518	12.12	0.0008
2	finewater		2	0.1587	0.2932	45.5945	17.29	<.0001
3	cementwater		3	0.1136	0.4069	28.3696	14.56	0.0003
4	slagwater		4	0.1339	0.5407	7.7178	21.86	<.0001

Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The REG Procedure
Model: MODEL1
Dependent Variable: compressivestrength



Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling
Input Data Set	CONCRETE100UP
Random Number Seed	777
Sampling Rate	0.8
Sample Size	92
Selection Probability	0.807018
Sampling Weight	0

Output Data Set	CONCRETE100UP_SAMPLE80
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Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The REG Procedure
Model: MODEL1
Dependent Variable: compressivestrength

Number of Observations Read	92
Number of Observations Used	92

Stepwise Selection: Step 1

Variable superplasticizerwater Entered: R-Square = 0.1092 and C(p) = 83.1980

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	927.99105	927.99105	11.03	0.0013
Error	90	7570.17784	84.11309		
Corrected Total	91	8498.16889			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	43.26595	1.16935	115151	1369.00	<.0001
superplasticizerwater	116.88873	35.19108	927.99105	11.03	0.0013

Bounds on condition number: 1, 1

Stepwise Selection: Step 2

Variable finewater Entered: R-Square = 0.2618 and C(p) = 55.8700

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	2224.84401	1112.42201	15.78	<.0001
Error	89	6273.32488	70.48680		
Corrected Total	91	8498.16889			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	64.20433	4.99747	11634	165.05	<.0001
superplasticizerwater	265.87595	47.37357	2220.20767	31.50	<.0001
finewater	-5.96769	1.39128	1296.85296	18.40	<.0001

Bounds on condition number: 2.1625, 8.6501

Stepwise Selection: Step 3

Variable cementwater Entered: R-Square = 0.3746 and C(p) = 36.1960

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	3183.24217	1061.08072	17.57	<.0001
Error	88	5314.92673	60.39689		
Corrected Total	91	8498.16889			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	50.76538	5.72548	4748.16033	78.62	<.0001
cementwater	7.58345	1.90371	958.39815	15.87	0.0001
superplasticizerwater	275.83984	43.92329	2381.98370	39.44	<.0001
finewater	-5.42353	1.29508	1059.21328	17.54	<.0001

Bounds on condition number: 2.1869, 16.221

Stepwise Selection: Step 4

Variable slagwater Entered: R-Square = 0.5279 and C(p) = 8.7385

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	4485.81670	1121.45418	24.32	<.0001
Error	87	4012.35219	46.11899		
Corrected Total	91	8498.16889			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	24.26730	7.06342	544.36757	11.80	0.0009
cementwater	15.02553	2.17447	2202.07807	47.75	<.0001
slagwater	16.91396	3.18261	1302.57453	28.24	<.0001

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
superplasticizerwater	256.18716	38.55972	2035.76091	44.14	<.0001
finewater	-2.28127	1.27684	147.21631	3.19	0.0775

Bounds on condition number: 2.7838, 34.935

Stepwise Selection: Step 5

Variable finewater Removed: R-Square = 0.5105 and C(p) = 10.0678

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	4338.60039	1446.20013	30.60	<.0001
Error	88	4159.56851	47.26782		
Corrected Total	91	8498.16889			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	13.43197	3.66590	634.57427	13.43	0.0004
cementwater	16.46194	2.04539	3061.78203	64.78	<.0001
slagwater	19.54705	2.85574	2214.57150	46.85	<.0001
superplasticizerwater	209.24981	28.57481	2534.71519	53.62	<.0001

Bounds on condition number: 1.5496, 12.8

Stepwise Selection: Step 6

Variable flyashwater Entered: R-Square = 0.5340 and C(p) = 7.5615

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	4537.86392	1134.46598	24.92	<.0001
Error	87	3960.30498	45.52075		
Corrected Total	91	8498.16889			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	9.60923	4.03491	258.17789	5.67	0.0194
cementwater	18.36783	2.20427	3160.78901	69.44	<.0001
slagwater	21.51132	2.95555	2411.39038	52.97	<.0001
flyashwater	9.32032	4.45473	199.26353	4.38	0.0393
superplasticizerwater	110.12872	55.05279	182.15953	4.00	0.0486

Bounds on condition number: 5.2776, 53.542

All variables left in the model are significant at the 0.0500 level.

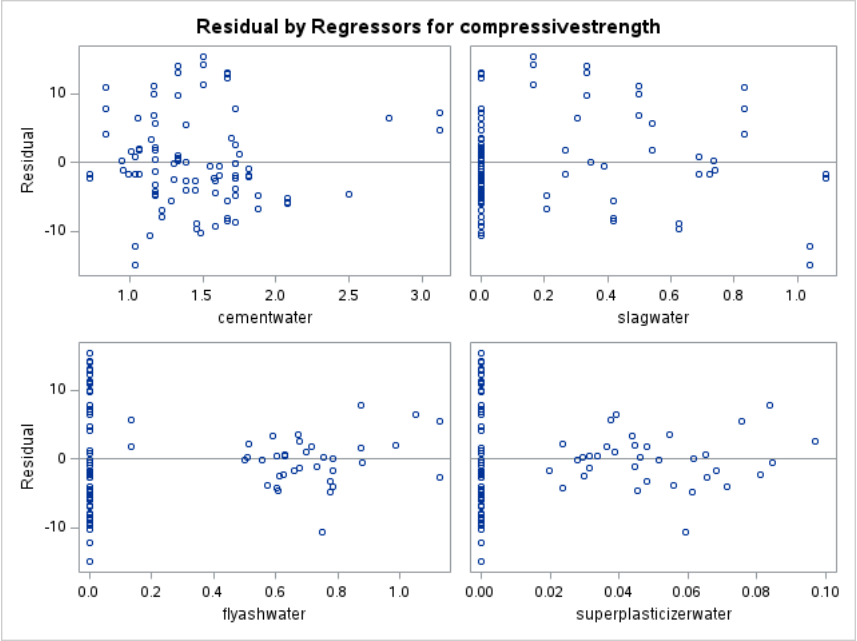
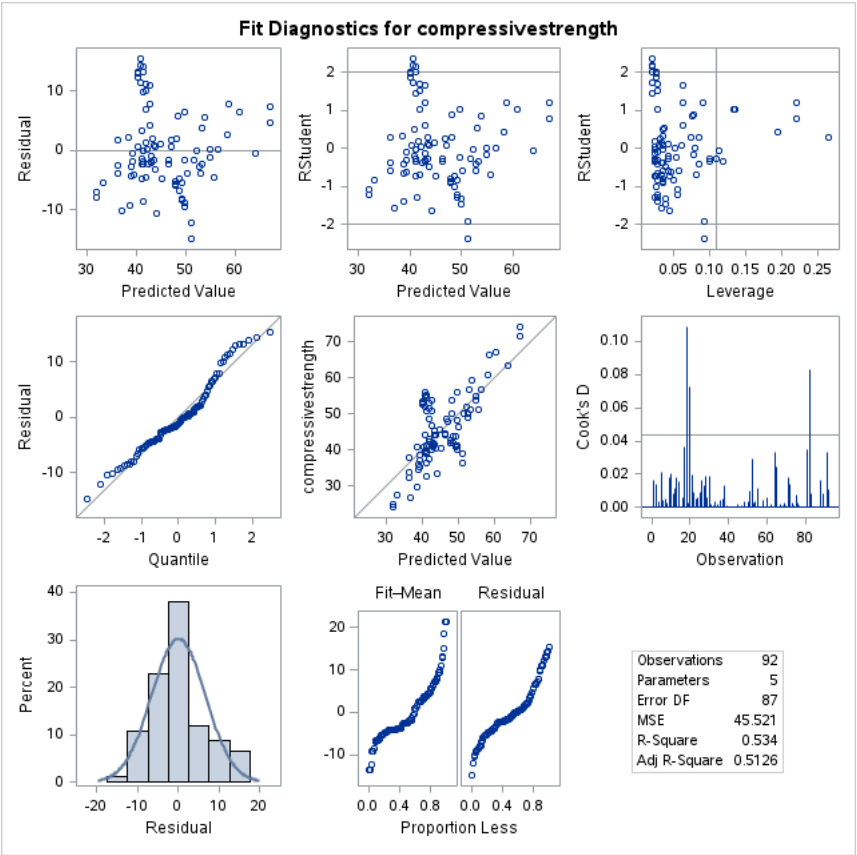
No other variable met the 0.0500 significance level for entry into the model.

Summary of Stepwise Selection								
Step	Variable Entered	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	superplasticizerwater		1	0.1092	0.1092	83.1980	11.03	0.0013
2	finewater		2	0.1526	0.2618	55.8700	18.40	<.0001
3	cementwater		3	0.1128	0.3746	36.1960	15.87	0.0001
4	slagwater		4	0.1533	0.5279	8.7385	28.24	<.0001
5		finewater	3	0.0173	0.5105	10.0678	3.19	0.0775
6	flyashwater		4	0.0234	0.5340	7.5615	4.38	0.0393

Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The REG Procedure
Model: MODEL1
Dependent Variable: compressivestrength



In both scenarios, cement-to-water ratio, slag-to-water ratio and superplasticizer-to-water ratio were consistently chosen, and the overall model fit remained alike ($R^2 = 0.541$ - 70 training split, $R^2 = 0.534$ - 80 training split, versus the full sample's: $R^2 = 0.557$). However, the fourth variable selected differed between the runs as fine-to-water ratio was selected in the 70 split, while fly ash-to-water was selected in the 80 training split. Fine-to-water ratio was actually removed partially through the run as well. It is known that stepwise selection has its limitations in terms of variable selection at times, so it is good practice to keep this in mind when interpreting which predictors are marked as most influential.

Note: Going forward, we will use only the 80/20 split data for simplicity.

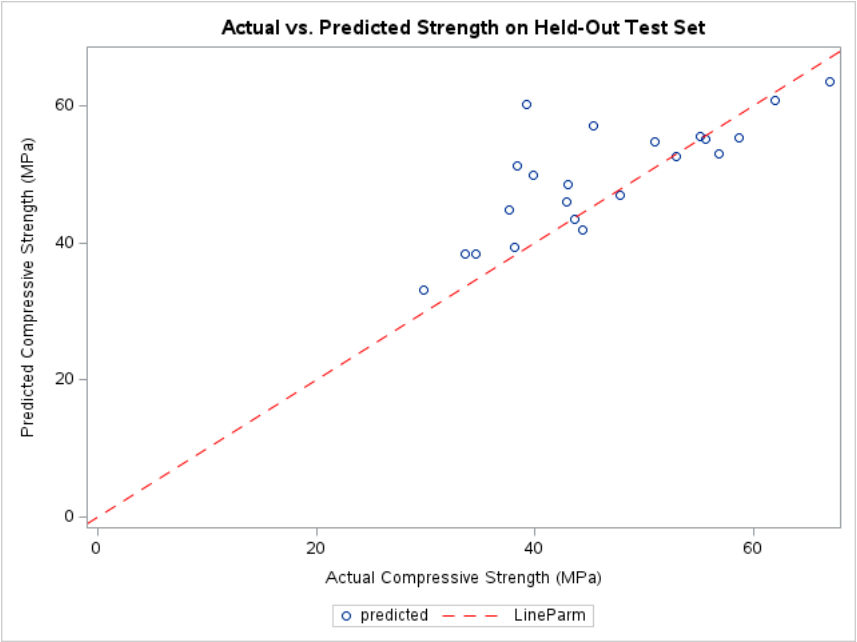
Mean Curing Age by Cluster

Section 3: Predictive Modeling I - Compressive Strength for Long-Cured Concrete

The REG Procedure
Model: MODEL1
Dependent Variable: compressivestrength

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	9.60923	4.03491	2.38	0.0194
cementwater	1	18.36783	2.20427	8.33	<.0001
slagwater	1	21.51132	2.95555	7.28	<.0001
superplasticizerwater	1	110.12872	55.05279	2.00	0.0486
flyashwater	1	9.32032	4.45473	2.09	0.0393



Actual vs. Predicted Strength on Held-Out Test Set

The MEANS Procedure

Analysis Variable : compressivestrength	
N	Mean
22	46.2308074

Actual vs. Predicted Strength on Held-Out Test Set

Out-of-Sample R-square
0.50006

In order to evaluate how well this model generalizes beyond the data it was originally fit on, it was tested on the withheld 20 test set. The out-of-sample R² was 0.500, which in comparison with 0.534 produced by the training set and 0.557 by the full set, depicts an expected decline. This is a good indicator that the model is not overfitting when using the training data and that it performs arguably well on new observations.

Based on this validation process, it is concluded that the recommended model for estimating compressive strength in concrete cured at least 100 days uses cement-to-water, slag-to-water, superplasticizer-to-water, and fly ash-to-water ratios as its predictors. This is consistent with the aforementioned Abrams' Law, with cement and superplasticizer ratios yielding the greatest positive effects on strength, with slag and fly ash contributing positively as well. And to recap, this model explains roughly half of the variability in strength on new data (out-of-sample R² = 0.500), providing a practical tool for estimating expected strength in older concrete based on its material composition. Notably, age did not enter any stepwise model, but this is to be expected given the sample is already restricted to concrete cured at least 100 days, which in turn leaves limited variation in age to detect a relationship.

For this task, the client requested an assessment of whether concrete cured within 90 and 100 days is likely to reach a compressive strength of 50MPa, a threshold which is important for reaching safety evaluation standards. Since the outcome is binary (i.e., reaching 50 MPa or not), logistic regression is appropriate, with the component-to-water ratios and age as predictors.

Within the 90-100 day curing period, 128 concrete samples were accessible, of which 51 (39.8) achieved the 50 MPa strength threshold and 77 (60.2) did not. This sample size is acceptable and supported for the use of logistic regression going forward.

Actual vs. Predicted Strength on Held-Out Test Set

Section 4: Predictive Modeling II - Safety Threshold Classification

The FREQ Procedure

high_strength	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	77	60.16	77	60.16
1	51	39.84	128	100.00

Before fitting the logistic regression models, predictor collinearity is assessed via a linear regression VIF diagnostic, since the variance inflation factors are not directly accessible when using the proc logistic call.

Since this is purely to view the variance inflation factors, we will omit discussion of the parameter estimates for now. All predictors showed variance inflation factors below 10 (ranging from 3.24 to 7.89), indicating no serious multicollinearity concerns within this subset. At this point, all component-to-water ratios and age were retained for the logistic models that follow.

Actual vs. Predicted Strength on Held-Out Test Set

Section 4: Predictive Modeling II - Safety Threshold Classification

The REG Procedure
Model: MODEL1
Dependent Variable: compressivestrength

Parameter Estimates						
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation
Intercept	1	-110.49539	26.66929	-4.14	<.0001	0
cementwater	1	20.53627	1.70636	12.04	<.0001	4.27540
slagwater	1	21.53641	1.76541	12.20	<.0001	2.47837
flyashwater	1	8.33046	3.80795	2.19	0.0306	7.88808
superplasticizerwater	1	-40.39346	28.99081	-1.39	0.1661	5.97070
coarsewater	1	0.54078	1.16169	0.47	0.6424	3.81749
finewater	1	-1.45411	1.02757	-1.42	0.1596	3.23621
age	1	1.28234	0.27230	4.71	<.0001	6.30984

A full logistic model was first fit using all component-to-water ratios as predictors (age withheld from this initial model). Global tests confirm that the model significantly improves prediction over an intercept-only model, and the odds ratios show that increases in key components increase the likelihood of reaching 50 MPa strength.

Actual vs. Predicted Strength on Held-Out Test Set

Section 4: Predictive Modeling II - Safety Threshold Classification

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CONCRETE90TO100
Response Variable	high_strength
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	174.127	66.400
SC	176.980	86.364
-2 Log L	172.127	52.400

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	119.7277	6	<.0001
Score	74.3184	6	<.0001
Wald	25.6965	6	0.0003

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-6.5056	3.4410	3.5744	0.0587
cementwater	1	7.6397	1.6882	20.4783	<.0001
slagwater	1	7.9216	1.7901	19.5834	<.0001
flyashwater	1	6.6410	2.5046	7.0304	0.0080
superplasticizerwate	1	88.5867	28.2608	9.8258	0.0017
coarsewater	1	-1.9027	0.9682	3.8617	0.0494
finewater	1	-0.6624	0.9176	0.5211	0.4704

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
cementwater	>999.999	76.010	>999.999
slagwater	>999.999	82.529	>999.999
flyashwater	765.886	5.652	>999.999
superplasticizerwater	>999.999	>999.999	>999.999
coarsewater	0.149	0.022	0.995
finewater	0.516	0.085	3.114

When looking at the outputs, an unusual pattern appears in the output: the odds ratios and confidence intervals yield some extremely large values. We can see that this occurs with the addition of superplasticizer water, given its extreme coefficient, but we shall assess this next.

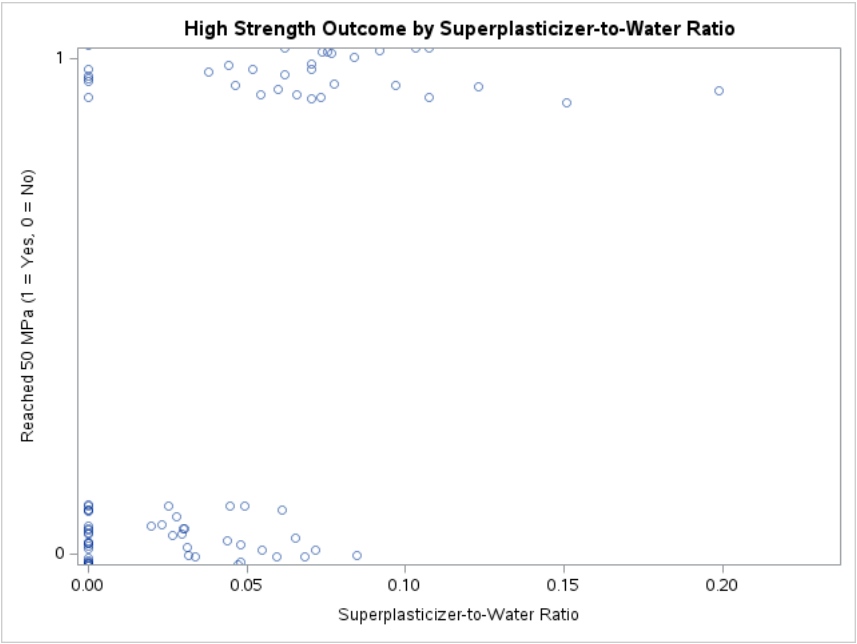
Before modeling, a cross-tabulation of high_strength versus superplasticizer-to-water ratio revealed a quasi-complete separation pattern (i.e., when one predictor almost perfectly predicts the outcome variable, with minimal overlap in the middle). At low values, basically all observations are high_strength = 0, and at high values, basically all are high_strength = 1, with minimal overlap in the mid-range. This explains the extremely large odds ratios and unstable confidence intervals initially observed.

Actual vs. Predicted Strength on Held-Out Test Set

Section 4: Predictive Modeling II - Safety Threshold Classification

The FREQ Procedure

Frequency Percent														
high_strength														
	0	0.0198399466	0.0230784779	0.0235617307	0.0236864992	0.02544486	0.0264097984	0.0277835696	0.0295059225	0.029802011	0.0305120722	0.0313988629	0.031556763	
0	46	1	1	1	1	1	1	1	1	1	1	1	1	
	35.94	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	
1	10	0	0	0	0	0	0	0	0	0	0	0	0	
	7.81	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
Total	56	1	1	1	1	1	1	1	1	1	1	1	1	
	43.75	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	0.78	



Here, stepwise selection is implemented, including age as an additional predictor.

High Strength Outcome by Superplasticizer-to-Water Ratio

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CONCRETE90TO100
Response Variable	high_strength
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	128
Number of Observations Used	128

Response Profile		
Ordered Value	high_strength	Total Frequency
1	0	77
2	1	51

Probability modeled is high_strength=1.

Stepwise Selection Procedure

Step 0. Intercept entered:

Model Convergence Status	
Convergence criterion (GCONV=1E-8) satisfied.	

-2 Log L	=	172.127
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Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
74.5323	7	<.0001

Step 1. Effect superplasticizerwate entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	174.127	126.767
SC	176.980	132.472
-2 Log L	172.127	122.767

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	49.3600	1	<.0001
Score	39.4759	1	<.0001
Wald	30.3747	1	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
48.6706	6	<.0001

Note: No effects for the model in Step 1 are removed.

Step 2. Effect cementwater entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	174.127	107.249
SC	176.980	115.805
-2 Log L	172.127	101.249

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	70.8783	2	<.0001
Score	49.2980	2	<.0001
Wald	30.6885	2	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
38.0901	5	<.0001

Note: No effects for the model in Step 2 are removed.

Step 3. Effect slagwater entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	174.127	73.731
SC	176.980	85.139
-2 Log L	172.127	65.731

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	106.3964	3	<.0001
Score	61.1653	3	<.0001
Wald	25.2923	3	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
12.0948	4	0.0167

Note: No effects for the model in Step 3 are removed.

Step 4. Effect finewater entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	174.127	71.410
SC	176.980	85.670

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
-2 Log L	172.127	61.410

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	110.7175	4	<.0001
Score	61.5954	4	<.0001
Wald	24.5708	4	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
8.8298	3	0.0316

Step 5. Effect finewater is removed:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	174.127	73.731
SC	176.980	85.139
-2 Log L	172.127	65.731

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	106.3964	3	<.0001
Score	61.1653	3	<.0001
Wald	25.2923	3	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
12.0948	4	0.0167

Note: No effects for the model in Step 5 are removed.

Note: Model building terminates because the last effect entered is removed by the Wald statistic criterion.

Summary of Stepwise Selection							
Step	Effect		DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq
	Entered	Removed					
1	superplasticizerwate		1	1	39.4759		<.0001
2	cementwater		1	2	22.0435		<.0001
3	slagwater		1	3	26.6605		<.0001
4	finewater		1	4	4.2477		0.0393
5		finewater	1	3		3.6925	0.0547

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-12.4253	2.4447	25.8331	<.0001
cementwater	1	5.1968	1.1403	20.7703	<.0001
slagwater	1	5.9010	1.3536	19.0052	<.0001
superplasticizerwate	1	76.9592	16.6005	21.4921	<.0001

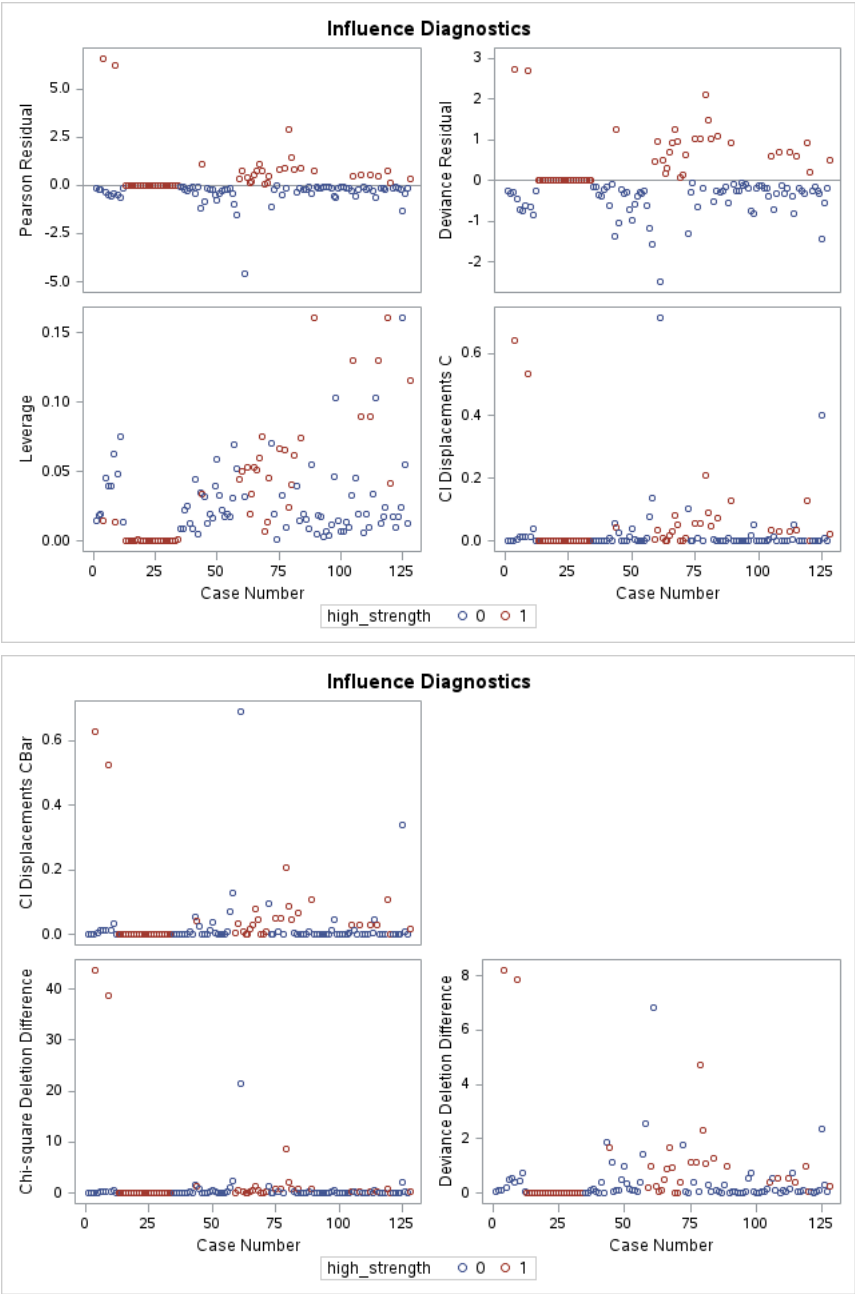
Odds Ratio Estimates		
Effect	Point Estimate	95% Wald Confidence Limits
cementwater	180.696	19.334 >999.999
slagwater	365.415	25.739 >999.999
superplasticizerwate	>999.999	>999.999 >999.999

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	95.1	Somers' D	0.903
Percent Discordant	4.8	Gamma	0.903
Percent Tied	0.1	Tau-a	0.436
Pairs	3927	c	0.951

Partition for the Hosmer and Lemeshow Test					
Group	Total	high_strength = 1		high_strength = 0	
		Observed	Expected	Observed	Expected
1	13	0	0.08	13	12.92
2	13	2	0.23	11	12.77
3	13	0	0.42	13	12.58
4	13	0	0.69	13	12.31
5	13	1	1.93	12	11.07

Partition for the Hosmer and Lemeshow Test					
Group	Total	high_strength = 1		high_strength = 0	
		Observed	Expected	Observed	Expected
6	13	3	4.30	10	8.70
7	13	9	7.98	4	5.02
8	13	12	11.37	1	1.63
9	13	13	12.98	0	0.02
10	11	11	11.00	0	0.00

Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
16.6316	8	0.0342



Influence diagnostics were reviewed in an effort to see if any individual observations were disproportionately controlling the model's results. A small number of observations showed modestly elevated influence, but none exceeded a Cook's Distance of 0.7, suggesting no single observation had a concerning impact on the model.

The Hosmer-Lemeshow goodness-of-fit test was statistically significant (Chi-square = 16.63, $p = 0.034$), indicating some evidence of some misalignment between the observed and predicted probabilities across risk groups. Despite this, the model's overall classification performance remained strong (concordance = 95.1), and this limitation is noted for transparency rather than treated as nullifying the model's usefulness.

Age did not enter the stepwise model at any step, due to that restricted 11-day curing window we discussed prior.

In an attempt to address this quasi-complete separation, Firth's penalized likelihood was used to correct for this; unlike standard maximum likelihood estimation, Firth's method adds a small penalty term that keeps coefficient estimates finite and stable even when a predictor nearly perfectly separates the outcome classes. However, when applied to the full six-predictor model, the odds ratios for several variables (particularly superplasticizer-to-water ratio) remained extreme. This indicates that fitting all six predictors simultaneously left too little independent information for the correction to fully stabilize each estimate. Because Firth's correction cannot be combined with automated stepwise selection in proc logistcs, the variable already identified as most influential by stepwise procedure (cement-to-water, slag-to-water, and superplasticizer-to-water ratios) were used to instead fit a smaller, more parsimonious Firth-corrected model. Reducing the number of predictors lessens the shared collinear signal driving the instability, and is expected to produce more stable, interpretable estimates than the full model.

High Strength Outcome by Superplasticizer-to-Water Ratio

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CONCRETE90TO100
Response Variable	high_strength
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring
Likelihood Penalty	Firth's bias correction

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	169.581	75.823
SC	172.433	87.232
-2 Log L	167.581	67.823

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	99.7572	3	<.0001
Score	61.5746	3	<.0001
Wald	25.6247	3	<.0001

Analysis of Penalized Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-11.3680	2.2168	26.2982	<.0001
cementwater	1	4.7314	1.0470	20.4193	<.0001
slagwater	1	5.3880	1.2421	18.8180	<.0001
superplasticizerwate	1	70.7543	15.2757	21.4537	<.0001

Odds Ratio Estimates		
Effect	Point Estimate	95% Wald Confidence Limits
cementwater	113.450	14.573 883.187
slagwater	218.766	19.175 >999.999
superplasticizerwate	>999.999	>999.999 >999.999

In order to stay consistent with previous procedures, we will split this data into a train/test split, in order to assess its ability to generalize on data beyond its original fit.

High Strength Outcome by Superplasticizer-to-Water Ratio

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling
------------------	------------------------

Input Data Set	CONCRETE90TO100
Random Number Seed	777
Sampling Rate	0.8
Sample Size	103
Selection Probability	0.804688
Sampling Weight	0
Output Data Set	CONCRETE90TO100_SAMPLE80

High Strength Outcome by Superplasticizer-to-Water Ratio

The LOGISTIC Procedure

Analysis of Penalized Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-10.1232	2.2661	19.9555	<.0001
cementwater	1	4.2726	1.1510	13.7802	0.0002
slagwater	1	4.7152	1.2238	14.8455	0.0001
superplasticizerwate	1	56.2495	14.7134	14.6154	0.0001

Since this is a model using logistic regression, we will need to calculate the scoring using a different method: the inverse-logit transformation. This is done to give us a valid probability between 0 to 1 that we can interpret.

Based on the confusion matrix, the withheld 20 test split correctly classified 22 of 25 observations (88.0 accuracy). The sensitivity (i.e., correctly identifying concrete that reached 50 MPa) was 81.8; the specificity (correctly identifying concrete that did not reach the threshold) was 92.9. This is a solid out-of-sample performance that backs the model's interpretability for classifying concrete samples within this curing period. However, it is good to note that the small test set size (n=25) means this should be used as a suggestive estimate rather than definitive.

High Strength Outcome by Superplasticizer-to-Water Ratio

The FREQ Procedure

Frequency	Table of high_strength by predicted_class			
high_strength	predicted_class			Total
	0	1		

Table of high_strength by predicted_class			
high_strength	predicted_class		
	0	1	Total
0	13	1	14
1	2	9	11
Total	15	10	25

All things considered, the reduced three-predictor Firth model, which uses cement-to-water, slag-to-water, and superplasticizer-to-water ratios, is recommended as the final tool for examining whether concrete cured within 90-100 days is likely to meet that 50 MPa safety standard. All three of these predictors remain strongly positive, though the exact magnitude of the superplasticizer effect could not be precisely assessed due to the separation discussed earlier. The model's strong out-of-sample performance (88.0 accuracy, 81.8 sensitivity, 92.9 specificity) supports its practical use, even with the small sample size. For the client, the greatest takeaway is that increasing cement, slag, and superplasticizer contents meaningfully increases the likelihood of reaching the 50 MPa safety threshold, given the curing window. However, precise probability estimates should be examined with some caution given both the separation issue and the limited sample size.

For our final task (3), we aim to develop a model that classifies concrete samples into one of the five age ranges specified based on their component-to-water ratios and compressive strength. The manager is interested in estimating the approximate age of concrete when the composition is fixed, which can help guide project timelines and quality control measures. To achieve this, we will use a logistic regression approach, allowing us to handle the multiple age categories and assess which concrete characteristics are most predictive of age. Model performance will be evaluated using classification accuracy and an analysis of which age groups are more difficult to distinguish

A linear regression of concrete component ratios, compressive strength and age explains 72 of the variability in age ($R^2 = 0.721$). Compressive strength is highly significant, while most component-to-water ratios are not, suggesting strength is the leading factor. Multicollinearity was not an apparent issue (all VIFs < 10).

A corrected linear regression predicting continuous age from component-to-water ratios and compressive strength explains approximately 34 of the variability in age ($R^2 \sim 0.343$). Age was excluded as a predictor since it would create a methodological circularity with the derived age-group variable used later in this section. This is substantially lower than the reported ($R^2 = 0.721$), which is a result of data leakage by using age to predict a variable directly derived from age (circular logic). The corrected model therefore provides a more appropriate estimate of predictive performance based solely on concrete composition and compressive strength.

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The REG Procedure
Model: MODEL1
Dependent Variable: agegroup

Root MSE	0.62484	R-Square	0.7207
Dependent Mean	2.20000	Adj R-Sq	0.7188
Coeff Var	28.40174		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1.29854	0.17566	7.39	<.0001
cementwater	1	-0.01426	0.04501	-0.32	0.7514
slagwater	1	0.05358	0.05611	0.95	0.3398
flyashwater	1	0.10056	0.08182	1.23	0.2193
superplasticizerwater	1	1.48401	0.87462	1.70	0.0901
coarsewater	1	-0.01120	0.03227	-0.35	0.7287
finewater	1	0.03234	0.03791	0.85	0.3939
age	1	0.01611	0.00032089	50.20	<.0001

Compressive strength is a statistically significant positive predictor of age ($t = 20.36$), ($p < 0.0001$), which is consistent with the assumption that concrete generally increases in strength as it cures over time. Cement-to-water, slag-to-water, and fly ash-to-water ratios were statistically significant as well, showing a negative association with age. On the other hand, superplasticizer-to-water and coarse-to-water ratios were not statistically significant. These relationships most likely show variation in mix-compositions for specific situations. Example, mixtures used for faster curing may differ structurally in composition from those created for longer curing periods. Variance inflation factors were also well below the 10 value threshold, with a maximum of 3.12; no multicollinearity concerns.

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The REG Procedure
Model: MODEL1
Dependent Variable: age

Root MSE	51.37794	R-Square	0.3430
Dependent Mean	45.66214	Adj R-Sq	0.3385
Coeff Var	112.51759		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	132.34896	13.91769	9.51	<.0001
cementwater	1	-53.62077	4.36565	-12.28	<.0001
slagwater	1	-58.43681	4.86879	-12.00	<.0001
flyashwater	1	-59.94201	6.81636	-8.79	<.0001
superplasticizerwater	1	60.14482	71.90616	0.84	0.4031
coarsewater	1	-3.09386	2.65281	-1.17	0.2438
finewater	1	-8.91493	3.10617	-2.87	0.0042
compressivestrength	1	2.61958	0.12869	20.36	<.0001

To further add to the claim of this circular logic, with the addition of Age as a predictor with Agegroup being our response variable, we notice that this triggers a complete separation of data points. This is a prime example of the data "leakage" discussed previously.

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CONCRETERATIOS
Response Variable	agegroup
Number of Response Levels	5
Model	cumulative logit
Optimization Technique	Fisher's scoring

Number of Observations Read	1030
Number of Observations Used	1030

Response Profile		
Ordered Value	agegroup	Total Frequency
1	1	324
2	2	425
3	3	91
4	4	131
5	5	59

Probabilities modeled are cumulated over the lower Ordered Values.

Stepwise Selection Procedure

Step 0. Intercepts entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

-2 Log L	=	2821.242
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Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
562.4738	7	<.0001

Step 1. Effect age entered:

Model Convergence Status
Complete separation of data points detected.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
5.2031	3	0.1575

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2829.242	20.482
SC	2848.991	45.169
-2 Log L	2821.242	10.482

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	2810.7597	1	<.0001
Score	542.0978	1	<.0001
Wald	92.9472	1	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
5.1581	6	0.5237

Note: No effects for the model in Step 1 are removed.

Step 2. Effect slagwater entered:

Model Convergence Status
Complete separation of data points detected.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
4.9322	6	0.5525

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2829.242	22.059
SC	2848.991	51.683

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
-2 Log L	2821.242	10.059

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	2811.1832	2	<.0001
Score	543.2306	2	<.0001
Wald	92.5431	2	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
4.8813	5	0.4305

Step 3. Effect slagwater is removed:

Model Convergence Status	
Complete separation of data points detected.	

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
5.2031	3	0.1575

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2829.242	20.482
SC	2848.991	45.169
-2 Log L	2821.242	10.482

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	2810.7597	1	<.0001
Score	542.0978	1	<.0001
Wald	92.9472	1	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
5.1581	6	0.5237

Note: No effects for the model in Step 3 are removed.

Note: Model building terminates because the last effect entered is removed by the Wald statistic criterion.

Summary of Stepwise Selection							
Step	Effect		DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq
	Entered	Removed					
1	age		1	1	542.0978		<.0001
2	slagwater		1	2	5.1499		0.0232
3		slagwater	1	1		1.0894	0.2966

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	1	11.6024	1.3684	71.8858	<.0001
Intercept	2	1	26.7431	4.3352	38.0554	<.0001
Intercept	3	1	40.5100	4.7979	71.2878	<.0001
Intercept	4	1	75.0519	7.9575	88.9557	<.0001
age		1	-0.6003	0.0623	92.9472	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
age	0.549	0.486	0.620

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	100.0	Somers' D	1.000
Percent Discordant	0.0	Gamma	1.000
Percent Tied	0.0	Tau-a	0.704
Pairs	373188	c	1.000

Partition for the Hosmer and Lemeshow Test											
Group	Total	Observed agegroup = 1	Observed agegroup = 2	Observed agegroup = 3	Observed agegroup = 4	Observed agegroup = 5	Expected agegroup = 1	Expected agegroup = 2	Expected agegroup = 3	Expected agegroup = 4	Expected agegroup = 5
1	136	136	0	0	0	0	136.0	0.01	0.00	0.00	0.00
2	126	126	0	0	0	0	125.9	0.08	0.00	0.00	0.00
3	62	62	0	0	0	0	59.57	2.43	0.00	0.00	0.00

Partition for the Hosmer and Lemeshow Test											
Group	Total	Observed agegroup = 1	Observed agegroup = 2	Observed agegroup = 3	Observed agegroup = 4	Observed agegroup = 5	Expected agegroup = 1	Expected agegroup = 2	Expected agegroup = 3	Expected agegroup = 4	Expected agegroup = 5
4	425	0	425	0	0	0	2.32	422.7	0.02	0.00	0.00
5	91	0	0	91	0	0	0.00	0.09	90.81	0.09	0.00
6	128	0	0	0	128	0	0.00	0.00	0.00	128.0	0.00
7	62	0	0	0	3	59	0.00	0.00	0.00	2.86	59.14

Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
5.1620	23	1.0000

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The LOGISTIC Procedure

Model Information	
Data Set	WORK.CONCRETERATIOS
Response Variable	agegroup
Number of Response Levels	5
Model	cumulative logit
Optimization Technique	Fisher's scoring

Number of Observations Read	1030
Number of Observations Used	1030

Response Profile		
Ordered Value	agegroup	Total Frequency
1	1	324
2	2	425
3	3	91
4	4	131
5	5	59

Probabilities modeled are cumulated over the lower Ordered Values.

Stepwise Selection Procedure

Step 0. Intercepts entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

-2 Log L	=	2821.242
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Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
635.3378	7	<.0001

Step 1. Effect compressivestrength entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
56.2818	3	<.0001

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2829.242	2484.611
SC	2848.991	2509.297
-2 Log L	2821.242	2474.611

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	346.6310	1	<.0001
Score	319.5974	1	<.0001
Wald	285.1848	1	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
444.0517	6	<.0001

Note: No effects for the model in Step 1 are removed.

Step 2. Effect cementwater entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
130.7946	6	<.0001

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2829.242	2226.993
SC	2848.991	2256.617
-2 Log L	2821.242	2214.993

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	606.2491	2	<.0001
Score	487.7831	2	<.0001
Wald	435.4729	2	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
276.8950	5	<.0001

Note: No effects for the model in Step 2 are removed.

Step 3. Effect slagwater entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
178.2933	9	<.0001

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2829.242	2074.495
SC	2848.991	2109.056
-2 Log L	2821.242	2060.495

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	760.7471	3	<.0001
Score	566.6619	3	<.0001
Wald	492.4577	3	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
166.8400	4	<.0001

Note: No effects for the model in Step 3 are removed.

Step 4. Effect flyashwater entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
271.5003	12	<.0001

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2829.242	1899.513
SC	2848.991	1939.012
-2 Log L	2821.242	1883.513

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	937.7288	4	<.0001
Score	627.7280	4	<.0001
Wald	535.3510	4	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
27.0908	3	<.0001

Note: No effects for the model in Step 4 are removed.

Step 5. Effect coarsewater entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
273.3137	15	<.0001

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2829.242	1884.850
SC	2848.991	1929.286
-2 Log L	2821.242	1866.850

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	954.3915	5	<.0001
Score	630.9652	5	<.0001
Wald	533.3444	5	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
11.9625	2	0.0025

Note: No effects for the model in Step 5 are removed.

Step 6. Effect superplasticizerwate entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Score Test for the Proportional Odds Assumption		
Chi-Square	DF	Pr > ChiSq
312.1029	18	<.0001

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2829.242	1875.963
SC	2848.991	1925.336
-2 Log L	2821.242	1855.963

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	965.2793	6	<.0001
Score	635.0896	6	<.0001
Wald	532.6388	6	<.0001

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
2.2029	1	0.1377

Note: No effects for the model in Step 6 are removed.

Note: No (additional) effects met the 0.05 significance level for entry into the model.

Summary of Stepwise Selection							
Step	Effect		DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq
	Entered	Removed					
1	compressivestrength		1	1	319.5974		<.0001
2	cementwater		1	2	232.2174		<.0001
3	slagwater		1	3	145.9116		<.0001
4	flyashwater		1	4	150.3061		<.0001
5	coarsewater		1	5	16.1963		<.0001
6	superplasticizerwate		1	6	9.4453		0.0021

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	1	-5.7790	0.5766	100.4369	<.0001
Intercept	2	1	-2.1508	0.5452	15.5650	<.0001

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	3	1	-1.2449	0.5462	5.1947	0.0227
Intercept	4	1	0.6638	0.5567	1.4217	0.2331
cementwater		1	4.4032	0.2463	319.7046	<.0001
slagwater		1	3.9373	0.2589	231.3102	<.0001
flyashwater		1	3.3878	0.3163	114.7028	<.0001
superplasticizerwater		1	-9.6153	2.7640	12.1014	0.0005
coarsewater		1	0.4636	0.0952	23.6991	<.0001
compressivestrength		1	-0.2097	0.00911	529.5621	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
cementwater	81.716	50.430	132.411
slagwater	51.280	30.874	85.174
flyashwater	29.601	15.924	55.024
superplasticizerwater	<0.001	<0.001	0.015
coarsewater	1.590	1.319	1.916
compressivestrength	0.811	0.796	0.825

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	88.9	Somers' D	0.780
Percent Discordant	10.9	Gamma	0.781
Percent Tied	0.2	Tau-a	0.549
Pairs	373188	c	0.890

Partition for the Hosmer and Lemeshow Test											
Group	Total	Observed agegroup = 1	Observed agegroup = 2	Observed agegroup = 3	Observed agegroup = 4	Observed agegroup = 5	Expected agegroup = 1	Expected agegroup = 2	Expected agegroup = 3	Expected agegroup = 4	Expected agegroup = 5
1	103	97	5	1	0	0	98.3	4.52	0.08	0.05	0.01
2	103	88	14	0	1	0	82.1	20.1	0.42	0.24	0.04
3	103	74	28	1	0	0	60.3	40.8	1.17	0.69	0.12
4	103	29	68	4	2	0	37.3	61.0	2.77	1.67	0.30
5	103	24	75	3	1	0	21.7	71.8	5.42	3.44	0.62
6	103	7	87	5	4	0	12.8	73.5	9.19	6.30	1.19
7	103	5	62	18	13	5	6.49	66.6	15.2	12.2	2.49
8	103	0	40	17	26	20	3.09	51.7	21.0	22.0	5.25
9	103	0	29	23	37	14	1.42	33.5	22.3	34.6	11.2
10	103	0	17	19	47	20	0.39	11.9	12.4	38.7	39.6

Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
113.2741	35	<.0001

After removal of age as a predictor, plus including compressive strength with the component-to-water ratios, cumulative logistic regression converged normally and yielded an acceptable discrimination (c = 0.890). Yet, two concerns still remain relevant which are 1. the Hosmer-Lemeshow test suggests a significant lack of fit ($\chi^2 = 113.27$, $p < 0.0001$) for especially higher age groups, and 2. the score test for the proportional odds assumption was significant at each step ($\chi^2 = 312.10$, $p < 0.0001$), suggesting that the assumption of each predictor's effect being consistent across all age-groups was potentially infringed. With these concerns in mind, discriminant analysis was explored as an alternative means that does not rely on the proportional odds assumption.

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The DISCRIM Procedure

Total Sample Size	1030	DF Total	1029
Variables	7	DF Within Classes	1025
Classes	5	DF Between Classes	4

Number of Observations Read	1030
Number of Observations Used	1030

Class Level Information					
agegroup	Variable Name	Frequency	Weight	Proportion	Prior Probability
1	1	324	324.0000	0.314563	0.200000
2	2	425	425.0000	0.412621	0.200000
3	3	91	91.0000	0.088350	0.200000
4	4	131	131.0000	0.127184	0.200000
5	5	59	59.0000	0.057282	0.200000

Within Covariance Matrix Information		
agegroup	Covariance Matrix Rank	Natural Log of the Determinant of the Covariance Matrix
1	7	-10.30850
2	7	-9.99257

Within Covariance Matrix Information		
agegroup	Covariance Matrix Rank	Natural Log of the Determinant of the Covariance Matrix
3	7	-11.58309
4	7	-11.24646
5	5	-51.09543
Pooled	7	-9.98953

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The DISCRIM Procedure
Test of Homogeneity of Within Covariance Matrices

Chi-Square	DF	Pr > ChiSq
2725.661599	112	<.0001

Since the Chi-Square value is significant at the 0.1 level, the within covariance matrices will be used in the discriminant function.
Reference: Morrison, D.F. (1976) Multivariate Statistical Methods p252.

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The DISCRIM Procedure

Generalized Squared Distance to agegroup					
From agegroup	1	2	3	4	5
1	-10.30850	-5.60308	3.92801	6.87772	218161983
2	-5.88638	-9.99257	-6.86260	-6.08877	311840244
3	1.11968	-7.16998	-11.58309	-9.99132	706816600
4	2.17521	-7.20291	-9.89684	-11.24646	228176541
5	3.62571	-4.96030	-2.20187	-9.35542	-51.09543

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The DISCRIM Procedure
Classification Summary for Calibration Data: WORK.CONCRETERATIOS
Resubstitution Summary using Quadratic Discriminant Function

Number of Observations and Percent Classified into agegroup						
From agegroup	1	2	3	4	5	Total
1	162 50.00	27 8.33	18 5.56	0 0.00	117 36.11	324 100.00
2	27 6.35	223 52.47	74 17.41	24 5.65	77 18.12	425 100.00
3	1 1.10	4 4.40	68 74.73	16 17.58	2 2.20	91 100.00
4	0 0.00	3 2.29	45 34.35	30 22.90	53 40.46	131 100.00
5	0 0.00	0 0.00	0 0.00	0 0.00	59 100.00	59 100.00
Total	190 18.45	257 24.95	205 19.90	70 6.80	308 29.90	1030 100.00
Priors	0.2	0.2	0.2	0.2	0.2	

Error Count Estimates for agegroup						
	1	2	3	4	5	Total
Rate	0.5000	0.4753	0.2527	0.7710	0.0000	0.3998
Priors	0.2000	0.2000	0.2000	0.2000	0.2000	

Before choosing between Linear and Quadratic discriminant analysis (LDA and QDA, respectively), we assessed whether the covariance structure of the predictors was the same across all five age groups. The results showed that they are not ($\chi^2 = 2725.66$, $df = 112$, $p < 0.0001$), which supports the use of QDA over LDA. QDA allows each group to its own covariance structure. Intriguingly, age group 5 stands out as its covariance matrix had a lower rank (5 versus 7 from the other groups), suggesting one or more predictors barely varied within that group.

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The MEANS Procedure

agegroup	N Obs	Variable	N	Mean	Std Dev
1	324	cementwater	324	1.652	0.646
		slagwater	324	0.375	0.485
		flyashwater	324	0.283	0.374
		superplasticizerwater	324	0.034	0.042
		coarsewater	324	5.609	0.758
		finewater	324	4.472	0.792
		compressivestrength	324	23.541	12.411

agegroup	N Obs	Variable	N	Mean	Std Dev
2	425	cementwater	425	1.480	0.654
		slagwater	425	0.470	0.478
		flyashwater	425	0.348	0.372
		superplasticizerwater	425	0.040	0.034
		coarsewater	425	5.296	0.830
		finewater	425	4.240	0.722
		compressivestrength	425	36.748	14.711
3	91	cementwater	91	1.782	0.662
		slagwater	91	0.339	0.441
		flyashwater	91	0.510	0.368
		superplasticizerwater	91	0.062	0.038
		coarsewater	91	5.916	0.777
		finewater	91	4.837	0.775
		compressivestrength	91	51.890	14.308
4	131	cementwater	131	1.567	0.624
		slagwater	131	0.390	0.463
		flyashwater	131	0.280	0.382
		superplasticizerwater	131	0.036	0.044
		coarsewater	131	5.533	0.866
		finewater	131	4.455	0.901
		compressivestrength	131	48.240	13.496
5	59	cementwater	59	1.591	0.532
		slagwater	59	0.270	0.356
		flyashwater	59	0.000	0.000
		superplasticizerwater	59	0.000	0.000
		coarsewater	59	4.666	0.676
		finewater	59	3.401	0.727
		compressivestrength	59	44.161	10.606

As previously suggested, age group 5's reduced covariance matrix rank was traced back to two predictors, fly ash-to-water ratio and superplasticizer-to-water ratio. These both yielded zero variance within the group (all 59 observations show a value of exactly 0 for each). This is an indicator that concrete mixes classified into this age group (cured 180+ days) consistently used zero fly ash or superplasticizer at all. This is a notable distinction of the composition compared to other age groups. This is a great tell of why QDA classifies group 5 with very high accuracy; the mixes are differentiable from the others based on composition alone, independent of curing time. This mirrors Cluster 4 from section 2, which consisted of long-cured, fly-ash free, high-cement mixes, indicating the two analyses are detecting the same grouping. Overall, the cross-validated misclassification rate is approximately 44, indicating that although the model captures general trends, there is significant overlap among groups 1, 2, and 4 in particular. This displays the limitations of the current selection of variables for classifying age with precision.

The quadratic discriminant analysis (QDA) performed on the dataset demonstrates a statistically significant ability to distinguish between the five age groups based on the component-to-water ratios and compressive strength. Multivariate tests, including Wilks' Lambda, Pillai's Trace, Hotelling-Lawley Trace, and Roy's Greatest Root, all indicate highly significant differences among the groups ($p < 0.0001$), confirming that the predictor variables collectively have distinguishing power. The cross-validation classification results show that QDA correctly classifies a substantial portion of observations in some age groups, particularly age groups 3 and 4, but struggles with groups 1, 2, and 5. The overall misclassification rate is approximately 44, suggesting that while the model captures some patterns in the data, certain age groups have overlapping characteristics, making them difficult to differentiate reliably. This outcome highlights why QDA is preferred over LDA in our case, the significant differences in the within-group covariance matrices violates the equal-covariance assumption needed for LDA, whereas QDA allows each age group to have its own covariance structure, improving the model's flexibility for classification.

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The DISCRIM Procedure

Multivariate Statistics and F Approximations					
S=4 M=1 N=508.5					
Statistic	Value	F Value	Num DF	Den DF	Pr > F
Wilks' Lambda	0.28990186	53.79	28	3675.5	<.0001
Pillai's Trace	0.86560989	40.32	28	4088	<.0001
Hotelling-Lawley Trace	1.94265734	70.62	28	2537.8	<.0001
Roy's Greatest Root	1.66654387	243.32	7	1022	<.0001
NOTE: F Statistic for Roy's Greatest Root is an upper bound.					

High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The DISCRIM Procedure

Classification Summary for Calibration Data: WORK.CONCRETERATIOS
Cross-validation Summary using Quadratic Discriminant Function

Number of Observations and Percent Classified into agegroup						
From agegroup	1	2	3	4	5	Total
1	160	27	20	0	117	324
	49.38	8.33	6.17	0.00	36.11	100.00
2	29	221	74	24	77	425
	6.82	52.00	17.41	5.65	18.12	100.00
3	1	7	57	24	2	91
	1.10	7.69	62.64	26.37	2.20	100.00
4	0	3	52	23	53	131
	0.00	2.29	39.69	17.56	40.46	100.00
5	0	0	0	0	59	59
	0.00	0.00	0.00	0.00	100.00	100.00
Total	190	258	203	71	308	1030
	18.45	25.05	19.71	6.89	29.90	100.00
Priors	0.2	0.2	0.2	0.2	0.2	

Error Count Estimates for agegroup						
	1	2	3	4	5	Total
Rate	0.5062	0.4800	0.3736	0.8244	0.0000	0.4368
Priors	0.2000	0.2000	0.2000	0.2000	0.2000	

Stepwise discriminant analysis was performed to identify the most important predictors for classifying the concrete age groups based on component-to-water ratios and compressive strength. Using a significance level of 0.05 for entry and stay, the procedure sequentially added predictors that most improved discrimination between the age groups. Compressive strength was the first variable entered, explaining the largest portion of variation (partial $R^2 = 0.338$, $F = 130.76$, $p < 0.0001$), indicating it is highly informative for distinguishing age groups. Cement-to-water ratio and slag-to-water ratio followed, both contributing significantly to group separation (partial $R^2 = 0.275$ and 0.154 , respectively, $p < 0.0001$). Subsequent variables—flyash-to-water ratio, fine water, superplasticizer-to-water ratio, and coarse water—also significantly improved classification, though their incremental contributions were smaller. Wilks' Lambda decreased with each step, confirming that each added variable enhances the model's ability to discriminate between groups. The average squared canonical correlation increased with each step, indicating that the model's overall predictive power improves as more relevant variables are included. This stepwise QDA model thus identifies the combination of component ratios and compressive strength that best distinguishes concrete age groups, with compressive strength being the dominant factor.

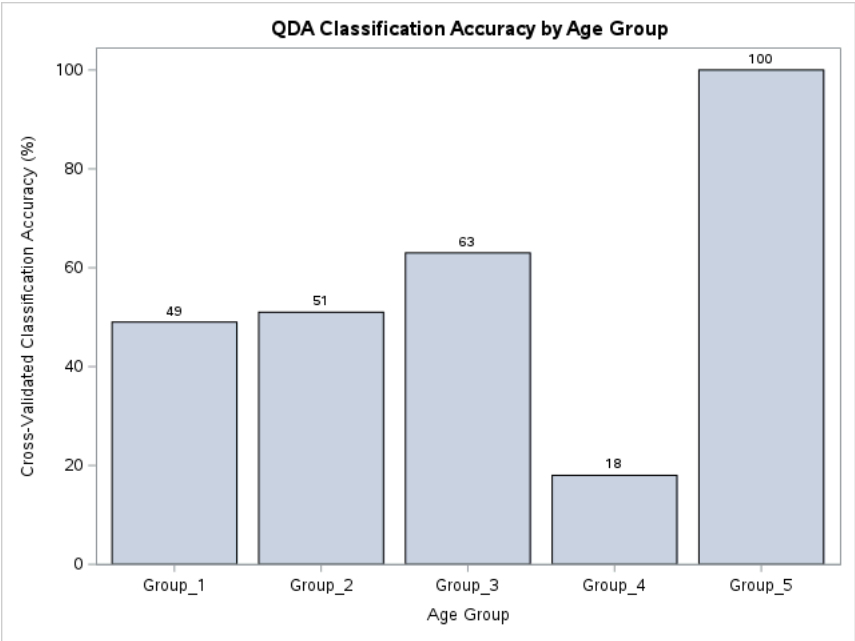
High Strength Outcome by Superplasticizer-to-Water Ratio

Section 5: Predictive Modeling III - Estimating Concrete Age

The STEPDISC Procedure

Stepwise Selection Summary										
Step	Number In	Entered	Removed	Partial R-Square	F Value	Pr > F	Wilks' Lambda	Pr < Lambda	Average Squared Canonical Correlation	Pr > ASCC
1	1	compressivestrength		0.3379	130.76	<.0001	0.66212192	<.0001	0.08446952	<.0001
2	2	cementwater		0.2753	97.23	<.0001	0.47987193	<.0001	0.13277208	<.0001
3	3	slagwater		0.1539	46.51	<.0001	0.40603300	<.0001	0.15331089	<.0001
4	4	flyashwater		0.1888	59.47	<.0001	0.32936483	<.0001	0.19038078	<.0001
5	5	finewater		0.0731	20.14	<.0001	0.30528034	<.0001	0.20754751	<.0001
6	6	superplasticizerwater		0.0323	8.52	<.0001	0.29540660	<.0001	0.21310871	<.0001
7	7	coarsewater		0.0186	4.84	0.0007	0.28990186	<.0001	0.21640247	<.0001

Using Quadratic Discriminant Analysis (QDA) with compressive strength and component-to-water ratios as predictors, we attempted to classify concrete samples into five age groups. The cross-validated classification results indicate that the model performs reasonably well for some groups but struggles with others. For instance, age groups 1, 2, and 3 show moderate classification accuracy, with 49, 51, and 63 of samples correctly classified, respectively. Age group 4 is more difficult to distinguish, with only 18 correctly classified. Age group 5 was classified correctly 100 of the time, due to the zero variance in fly ash-to-water and superplasticizer-to-water ratios within this group. This suggests that while the model captures general trends, there are significant overlaps between certain age groups. This indicates that age group 4 is particularly challenging to differentiate using these predictors, displaying the limits of the current set of variables for precise age classification.



This analysis examined how concrete analysis confirmed that composition and curing age relate to compressive strength, and developed a set of predictive models to support the client's decision-making around mixture design, safety thresholds, and quality control.

Descriptive and correlation analysis proved that cement content and curing age are the two greatest drivers of compressive strength, which is consistent with Abrams' Law. The predictor group overall did not show any major signs of multicollinearity concerns. Clustering pinpointed four distinct mix profiles, most notably a high-cement, high-slag, low-fly-ash, mixture (Cluster 1) which achieved strength akin to the longest-cured mixtures at a fraction of the curing age.

With regards to predicting the compressive strength of concrete cured at least 100 days, a validated stepwise regression model using cement, slag, superplasticizer, and fly ash ratios explained about half of the variability in strength on new, unseen data (out-of-sample R² = 0.500). This leaves us with a practical tool for estimating the expected strength in older concrete.

For checking whether concrete cured within a 90-100 days window reaches the 50 MPa safety threshold, an initial logistic model revealed quasi-complete separation driven by superplasticizer-to-water ratio. After applying Firth's penalized likelihood correction and validating the held-out test set, the final three-predictor model achieved a strong out-of-sample performance (88.0 accuracy), supporting cement, slag, and superplasticizer content as key components of reaching the strength threshold. However, the probability estimates should be interpreted with slight caution given the separation detected.

For estimating concrete age, an initial regression revealed data leakage from having agegroup as the response alongside age as a predictor variable. Once corrected, composition and strength explained a more modest but valid 34 of age's variability. A parallel attempt at logistic classification of age groups confirmed the same leakage issue through complete separation. After removing age, it showed reasonable discrimination but still violated a key modeling assumption. Quadratic discriminant analysis was then chosen since it does not rely on that assumption, and it classified age groups with an overall accuracy of roughly 56 (44 misclassification), performing the best for age groups 3 and 4 and most reliably for age group 5. Age group 5's distinct composition consisting of fly-ash-free and long-cured quality falls in line with Cluster 4 from the clustering analysis section, also making it more easily distinguishable from other groups.

To conclude everything, these analyses confirm that both mixture composition and curing time purposefully influence concrete performance, despite their relative importance varying by task. Composition dominates in differentiating high-performing mixes and safety thresholds, while age remains only somewhat predictable based off composition and strength alone. Many models required correcting for methodological issues identified during analysis, including data leakage, quasi-complete separation, and violated model assumptions, all of which emphasize the importance of diagnostic validation before choosing any given model's results. The corrected, validated models presented here provide the client with a practical, evidential foundation for mixture design, safety evaluation and quality control decisions.